



Socio-spatial equity in public transit system accessibility for slum and non-slum population in a developing mega city

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ABSTRACT

Inequity in public transport system access affects the mobility and overall social and economic development of vulnerable groups, who rely on public transport for accessing jobs, healthcare, and education. Very few studies discuss the interplay between accessing the transit system and social vulnerability, especially in developing cities. This study presents a comprehensive framework for measuring social and spatial inequity in transit accessibility by integrating Public Transport Accessibility Levels (PTAL), transit gap, and social vulnerability. PTAL measures a transit supply component, while the potential transit-dependent population is the demand. The difference between transit demand and transit supply is the transit gap. The Gini index is used to quantify the degree of accessibility, while bivariate analysis is used to explore spatial patterns and critical areas where transit accessibility is most problematic. Additionally, a Social Vulnerability Index is designed based on socioeconomic indicators, which helps to address public transit needs beyond slum and non-slum classifications. PTAL analysis found that 31.8% of Mumbai's population falls below the Good transit accessibility level. 52.2% of the population lives in high transit gap areas, and 13.6% live in transit deserts. It is observed that the accessibility to transit is not equitable for different population classes. For example, 32.5% of the highly vulnerable population is in low accessibility-high gap regions. Based on PTAL, transit gap, and social vulnerability, the study suggests targeted policy measures for reducing socio-spatial inequities in transit system accessibility. These findings can aid policymakers in improving transit accessibility and equity for vulnerable populations.

1. Introduction

Urbanization refers to the growth in urban population and expansion of urban areas, driven by commercialization, industrialization, migration, and modernization (Spence et al., 2008). As of 2018, 55% of the global population resides in urban areas, a figure expected to rise to 68% by the year 2050. India, China, and Nigeria alone are expected to contribute to 35% of the global urban population increase between 2018 and 2050 (UN DESA, 2019). In 2001, India had 35 cities with a population of one million or more, which increased to 53 by 2011, illustrating a rapid urban expansion (Census of India, 2011). This rapid urbanization has spawned a host of urban challenges, notably spatial imbalances in the supply and demand for public transport (Jiao & Dillivan, 2013). The spatial

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imbalance is recognised as one of the critical factors that contribute to urban social inequity (Lee et al., 2021; Sharma & Patil, 2021). Reducing inequity in urban settlements is challenging. The United Nations Department of Economic and Social Affairs (UN DESA) has proposed four key strategies to address this issue, one of which is providing a well-planned and efficient transport system to enhance accessibility to essential services (UN DESA, 2020).

One of the fundamental goals of public transit is to benefit the public, irrespective of their social, spatial, and financial status (Federal Transit Administration, 2012). Unfortunately, public transit authorities mostly work on providing similar transit services across the city, regardless of the need and demand (Carleton & Porter, 2018). Many previous studies have highlighted the inequitable distribution of public transit accessibility, particularly for marginalized and disadvantaged communities. For example, Ermagun & Tilahun (2020) examined accessibility to destinations such as jobs, schools, and healthcare across different socioeconomic groups in Chicago, USA, and found that low-income and minority populations, especially Black and Hispanic residents, face inequitably lower levels of access to reliable and frequent public transport compared to higher-income and white populations. In Wuhan, China, Cai et al. (2020) found that lower-income commuters and migrant populations tend to reside in suburban areas, where both physical access to transit stops and transit service frequency are limited, reducing their chances of connecting to high job-density business districts. Similarly, Lee et al. (2021) for Seoul, South Korea, noted that outer urban, low-density areas with a higher concentration of elderly residents experienced lower levels of transit accessibility compared to the rest of the city, due to limited public transit supply, even when infrastructure is already in place. A study by Sharma & Patil (2021) on accessibility to public healthcare centres in Mumbai shows that slum residents face poorer public transit accessibility to healthcare centres. The authors observed similar results for accessibility to schools using public transport (Sharma & Patil, 2022). Furthermore, financial constraints and physical disabilities can limit individuals' ability to access transport systems and opportunities, preventing them from participating in economic and social activities (Cai et al., 2020; Mohri et al., 2021; Tahmasbi et al., 2019). These findings emphasize that transit accessibility is not just about infrastructure; it is closely tied to social, spatial, and economic factors that can systematically disadvantage certain groups.

Counting only transit stops without considering route availability and frequency does not provide a complete picture of transit supply (Aman & Smith-Colin, 2020). Access to the transit system is a fundamental precondition for broader accessibility to employment, healthcare, education, leisure, social, and other opportunities (Aman & Smith-Colin, 2020; Cai et al., 2020; Jiao & Dillivan, 2013; Lee et al., 2021). Many transit studies focused on the access by the transit system to the opportunities, and very few analysed access to the transit system. Additionally, access to the transit system was indirectly measured using the number of stops and train (metro and suburban rail) stations in the vicinity (Sharma & Patil, 2021). Other important factors, such as route availability, service frequency, and integration across modes, that are essential to the actual usability of public transit, were ignored (Aman & Smith-Colin, 2020). Most of the studies on transit accessibility equity issues are from developed cities, where social vulnerability is often unidimensional, such as income inequality, an ageing population, or physical disabilities (Carleton & Porter, 2018; El-Geneidy et al., 2016; Ermagun & Tilahun, 2020; Mohri et al., 2021; Nie et al., 2024). The developing cities face more complex and layered forms of vulnerability that stem not only from income inequities but also from deficits in housing and hygiene quality, financial exclusion, and informal settlements (Bardhan et al., 2015; Roy et al., 2023; Sharma & Patil, 2022). As a result, measurement approaches and policy responses in developing cities must adopt a multidimensional approach.

This study proposes a comprehensive framework integrating access to the transit system, transit gap (defined as the mismatch between transit supply and potential transit demand), and social vulnerability using Mumbai as a case study. The following section reviews literature on accessibility, transit gaps, and transit equity, followed by an overview of the study area and data sources. The methodology outlines PTAL and transit gap estimation, equity assessment methods, and the approach for developing a social vulnerability index. The subsequent section details the integrated PTAL–transit gap–social vulnerability framework and related policy interventions to reduce accessibility inequity. The paper concludes with key findings and future research directions.

2. Literature review

This literature review segment is divided into three sections. The initial section delves into the different accessibility definitions and their several measurement approaches. The following sections examine the definition of the transit desert and methods for equity assessment, offering insights into how these aspects are interpreted and evaluated across the fields of transportation planning and urban development.

2.1. Accessibility measure

Accessibility is a multifaceted and interdisciplinary concept utilized in many fields, such as urban studies, transport planning, geography, social studies (Geurs & Van Wee, 2004), and information technology (Botelho, 2021). There is no unique definition for accessibility; it changes with specific focus and applications. In general, accessibility refers to the ease of getting goods, amenities, services, or reaching destinations. In transportation, Hansen (1959) pioneered the concept of accessibility and defined it as “the potential of opportunities for interaction.” According to Dalvi and Martin (1976), accessibility in transportation planning refers to “the ease with which any land-use activity can be reached from a location using a particular transport system.” There are a few other studies concentrating on specific categories, such as individual accessibility (Jin et al., 2022), spatial accessibility (area accessibility) (Chang et al., 2019; Mohri et al., 2021; Sharma & Patil, 2021), and the economic implications of accessibility (Cascetta et al., 2020).

Transit accessibility encompasses two parts: system accessibility and system-facilitated accessibility (Malekzadeh & Chung, 2020). Transit system accessibility addresses the ease of accessing public transport networks, such as transit stop numbers, connectivity, service frequency, and walkability to transit stations. Transit system facilitated accessibility evaluates a traveller's ability to reach

opportunities by factoring in the travel costs within the transit network (Malekzadeh & Chung, 2020). System-facilitated accessibility comprises two main components: the location of urban opportunity and its attractiveness, and travel cost (impedance factor) (Ermagun & Tilahun, 2020; Hansen, 1959; Sharma & Patil, 2021). Multiple factors, including the proximity to transit stops, service frequency, coverage of routes, and service reliability, influence transit system accessibility (Murray & Wu, 2003). These factors significantly influence travel behaviour and user satisfaction. Improved transit coverage, frequent service availability, and enhanced service reliability contribute to increased transit ridership (El-Geneidy et al., 2016).

Various methods have been developed to evaluate transit system accessibility, each with its own advancements and limitations. The common approach to assessing public transit system accessibility involves identifying the number of stops and routes within a specified radius. Foda and Osman (2010) utilized distance-based measures to count bus stops, such as the Ideal Stop Accessibility Indicators (ISAI) and Actual Stop Accessibility Indicators (ASAI). ISAI is based on the fixed coverage area (400 m radius buffer), and ASAI considers the coverage area based on 400 m access through actual pedestrian length. However, ISAI and ASAI values can be misleading, as lower scores may indicate wider coverage areas rather than reduced accessibility. Cheng and Agrawal (2010) proposed the Time-Based Transit Service Area Tool (TTSAT), which uses transit network data (road network, stop location, maximum walk distance, and in-transit travel time) and all possible destinations. TTSAT produces polygons showing areas covered within a defined time threshold. Pitot et al. (2006) developed the Land Use & Public Transport Accessibility Index (LUPTAI), a tool designed to evaluate how accessible various locations are to public transport stops, considering their surrounding land use patterns (such as residential, commercial, and recreational areas). This index integrates land use characteristics with the public transport network. It considers the concentration of different land use types around different transit mode stations. The Public Transport Accessibility Level (PTAL) evaluates system accessibility by considering factors such as proximity to stops, wait times, service frequency, reliability, and availability of various transit modes (Adhvaryu et al., 2019; Saghapour et al., 2016). Compared to tools like TTSAT and LUPTAI, PTAL demands less information and computational power (Adhvaryu & Kumar, 2021), making it suitable for understanding the system accessibility with limited datasets.

The PTAL method was originally developed for the London Borough of Hammersmith and Fulham in 1992 (Transport for London, 2010) and later adopted across London to support transport planning (Transport for London, 2015). It is widely applied in developed and developing cities to assess accessibility to transit stops. In developed contexts, cities such as Auckland (Wu & Hine, 2003), Melbourne (Saghapour et al., 2016), and Singapore (Yadav et al., 2024) have utilized the PTAL method. Similarly, in developing cities such as Ahmedabad (Shah & Adhvaryu, 2016), Surat (Adhvaryu et al., 2019), and Lucknow (Adhvaryu & Kumar, 2021), PTAL has proven to be a scalable and adaptable tool. It is also used to understand transit equity. Nie et al. (2024) employed PTAL to explore disparities in public transport accessibility across socioeconomic groups in London. Adhvaryu & Kumar (2021) also used PTAL maps in Lucknow, India, to examine variations in accessibility levels among slum dwellers and economically weaker sections.

2.2. Transit gap and transit desert

Previous research on transit accessibility has often focused either on the demand or supply side of transit systems, typically lacking a comprehensive perspective. Jiao and Dillivan (2013) proposed the idea of ‘transit desert’, taking inspiration from ‘food desert’. An area can be referred to as a food desert when access to nutritious and affordable food is limited, either due to the lack of grocery stores or food markets within a walkable distance, or inadequate transit infrastructure (Abel & Faust, 2020). Similarly, transit deserts are areas where the availability of transit services significantly falls short of meeting the demand for such services (Aman & Smith-Colin, 2020; Cai et al., 2020; Chen & Jiao, 2022; Jiao, 2017; Lee et al., 2021). Transit desert research focuses on three primary concepts: transit demand, transit supply, and the transit gap. Estimates of transit demand typically include populations that heavily rely on transit services, such as individuals with physical disabilities, the elderly, students, and young people who do not have a driving licence, and economically disadvantaged groups who may not have access to personal vehicles (Aman & Smith-Colin, 2020; Chen & Jiao, 2022; Lee et al., 2021). The transit gap highlights the disparity between the supply of transit services and transit demand (Cai et al., 2020; Lee et al., 2021). When this gap becomes substantial, a transit desert emerges, leaving parts of the population with inadequate transit access. Individuals staying in a transit desert encounter significant difficulties in accessing education, healthcare, employment, and engaging in leisure or social activities. Inadequate and unreliable public transportation services exacerbate these difficulties (Aman & Smith-Colin, 2020).

The transit desert method has been applied and tested globally, including in Seoul (Lee et al., 2021), Dallas (Aman & Smith-Colin, 2020), Wuhan (Cai et al., 2020), Texas (Jiao, 2017), and several European cities (Chen & Jiao, 2022). Yadav et al. (2024) argue against defining transit desert solely based on a high transit gap. They propose that transit deserts should be redefined as areas where transit supply is unsatisfactory, and demand generated is high. The areas with higher transit supply but overwhelming demand are better classified as transit-stressed areas rather than transit deserts. This study adopts a similar definition and methodology for analyzing transit deserts.

2.3. Equity assessment

Accessibility can also be employed as an indicator to estimate social and spatial equity in the context of transportation planning (Sclar et al., 2014). Equality and equity are often used interchangeably in transportation studies, but have somewhat different meanings and implications (Aman & Smith-Colin, 2020). Equality in transportation accessibility implies that all individuals have the same level of access to transportation services, irrespective of their socioeconomic status, recipient needs, and condition (Van Wee & Geurs, 2011). Equity, conversely, ensures that transportation services are distributed in a way that addresses the specific needs of

disadvantaged groups and reduces disparities (El-Geneidy et al., 2016; Ermagun & Tilahun, 2020; Pereira et al., 2017). Equitable distribution of transportation resources is critical because transportation is not merely about mobility, but also about access to opportunities, services, and activities that contribute to a high quality of life (Pereira et al., 2017). The difficulty in transportation access can limit opportunities for marginalised groups to access employment (Cai et al., 2020; Ermagun & Tilahun, 2020), education (Sharma & Patil, 2022), healthcare (Jin et al., 2022; Sharma & Patil, 2021), park (Chang et al., 2019), and exacerbate social inequities (Ermagun & Tilahun, 2020; Tahmasbi et al., 2019). Promoting public transportation equity aligns with the United Nations Sustainable Development Goals (SDGs), particularly those aimed at reducing poverty (SDG 1), ensuring inclusive and equitable quality education (SDG 4), and promoting sustained, inclusive, and sustainable economic growth (SDG 11) (UN, 2015). Multiple studies explore the role of equity in transportation planning, particularly regarding accessibility. However, there is no consensus on a single definition of equity in this context, possibly due to various ways of measuring accessibility.

Equity can be divided into two categories: horizontal equity and vertical equity. From the view of transportation studies, horizontal equity (equivalent to equality) emphasizes fairness by distributing resources evenly among socioeconomic groups with similar abilities and needs. These resources can include income, wealth, education, healthcare, and access to infrastructure. In this study, we consider access to transportation stops and the frequency of services available at those stops as key resources, reflecting individuals' ability to participate in urban opportunities through the public transport network. In transportation studies, horizontal equity is used to assess the distribution of transit infrastructure across various city regions, particularly in the context of uneven public transport allocation (Carleton & Porter, 2018). Vertical equity, on the other hand, focuses on allocating resources based on the differing abilities and needs of various socioeconomic groups (Aman & Smith-Colin, 2020; Ermagun & Tilahun, 2020; Mohri et al., 2021). In a complex and socially diverse city like Mumbai, where in 2011, 42.2% of the total population (Census of India, 2011) resided in slum areas with poor infrastructure, vertical equity becomes particularly important to ensure that those with greater needs receive proportionately greater transit access. Analysing transportation equity is challenging due to the interplay of multiple equity types together (i.e., horizontal and vertical equity), the need to account for various socioeconomic groups, and the evaluation of several impacts, including public transport facilities and services, economic opportunities, and user costs and benefits (Carleton & Porter, 2018).

Equity can also be classified into spatial equity and social equity. Resource distribution among regions is defined as spatial equity, and that among population groups is defined as social equity. The population can be grouped based on one or a combination of socioeconomic, demographic, housing and hygiene, education, health status, and infrastructure-related factors (see Table 2). Proximity to public transport, distance, and travel time to essential services are considered for spatial equity (Chang et al., 2019; Foda & Osman, 2010; Shah & Adhvaryu, 2016), and income level, educational attainment, age groups, and gender consideration for social equity (Sharma & Patil, 2021; Van Wee & Geurs, 2011). Studying both types of equities is critical because effective urban planning requires public transit access to all regions and all income groups. This paper has analysed both spatial and social equities in Mumbai city.

Social equity can be assessed by studying transit accessibility for socially vulnerable populations. Social vulnerability refers to the

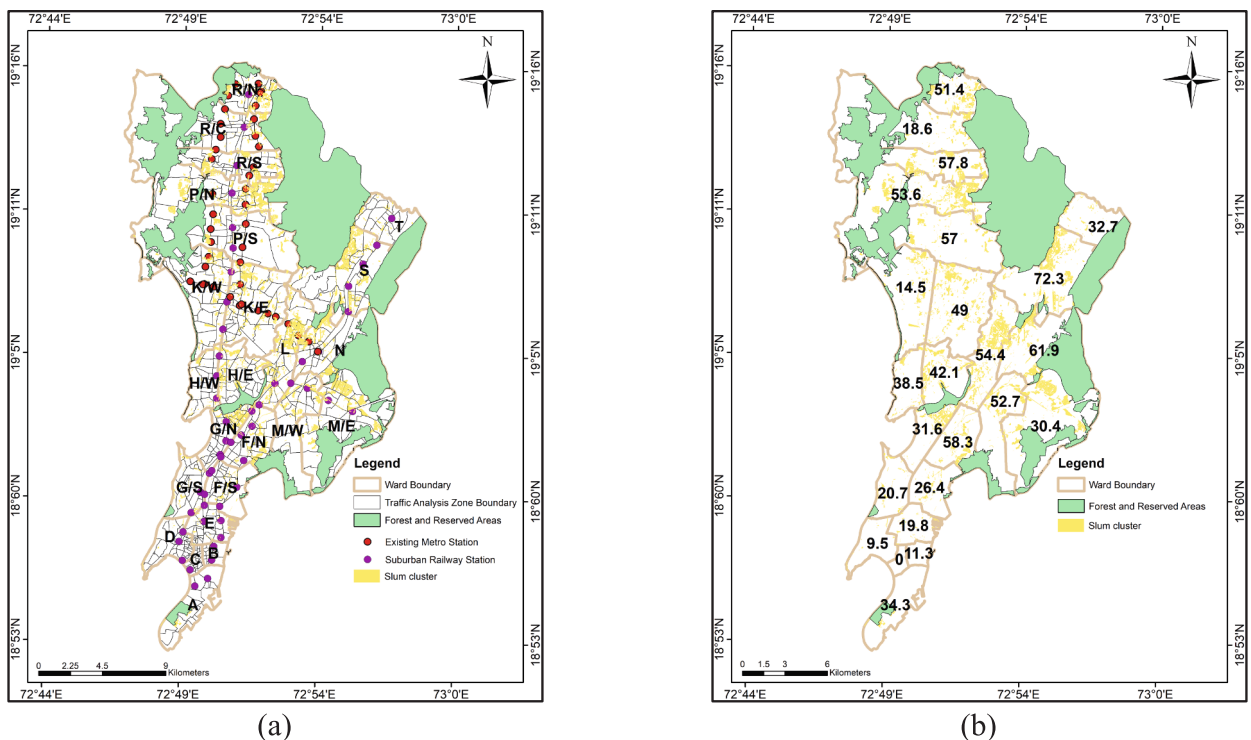


Fig. 1. (a) Distribution of suburban train and metro stations in Mumbai (b) Percentage of slum population in each ward of Mumbai.

susceptibility of certain population groups to harm or adverse effects due to various socioeconomic and environmental factors (Maiti et al., 2017). It encompasses a wide range of elements that can affect an individual's or a community's ability to prepare for, respond to, and recover from adverse situations, such as inadequate access to essential services, discrimination, economic downturns, social exclusion, and natural disasters. The multidimensional nature of vulnerability requires a holistic approach to address it effectively (Majumder et al., 2023). In the urban context, social vulnerability is framed by three main components: exposure, sensitivity, and adaptive capacity. Exposure refers to the degree to which urban populations are subjected to hazards such as natural disasters or socioeconomic stresses. Sensitivity denotes how susceptible these populations are to harm, influenced by factors such as age, health, and socioeconomic status. Adaptive capacity is the ability of urban communities to withstand, recover from, and adapt to adverse conditions determined by access to resources, social networks, and infrastructure (Maiti et al., 2017; Majumder et al., 2023).

Recent literature primarily focuses on transit system-facilitated accessibility, which refers to access to various opportunities through the transit system. However, there is limited attention to transit system accessibility, encompassing access to transit stops and the number of transit routes and frequency provided at these stops. Evaluating transit system accessibility is crucial, as it is the gateway to transit system-facilitated accessibility. Good system accessibility enables individuals to reach vital goods, opportunities, and activities, such as food, healthcare, education, and employment (Van Wee & Geurs, 2011). This research tries to develop a comprehensive framework that integrates access to the transit system, transit gap (defined by the mismatch between transit supply and potential transit demand), and social vulnerability. Access to public transit is estimated using the Public Transport Accessibility Level (PTAL) method. PTAL values represent the level of transit supply, and the difference between this supply and estimated potential transit demand is used to calculate the transit gap. By analysing PTAL, transit gap, and social vulnerability, this study aims to highlight spatial and social disparities in public transit access across Mumbai. Our analysis targets wards (administrative divisions) and traffic analysis zones (TAZs) of Mumbai and suggests appropriate policy recommendations. Identifying locations with insufficient transit supply and high demand and developing policies based on the existing infrastructure are critical steps toward reducing socio-spatial inequities in transit system accessibility.

3. Study area and datasets

Mumbai city, the study area, has attracted millions of migrants seeking business and service opportunities. Most of these migrants live in slum areas spread across the city. Fig. 1(a) depicts the suburban railway and metro stations and slum clusters of Mumbai. With a population of 12.4 million (Census of India, 2011), Mumbai covers an area of 457.5 square kilometres and is divided into 24 municipal wards for administrative purposes. As per the Comprehensive Transportation Study (CTS, 2019), Mumbai is further segmented into 640 traffic analysis zones (TAZs). Census data from 2011 (Census of India, 2011) indicates that 42.2% of the population lived in slum clusters; 11 out of 24 municipal wards have over 40% of their population residing in slums (Fig. 1(b)). Mumbai's slums are densely populated; for example, Dharavi, one of the densest slums in the world, has a population density of approximately 340 thousand people per square kilometre (Kaushal & Mahajan, 2021).

The Brihanmumbai Electric Supply and Transport (BEST) operates a public bus service in Mumbai, managing a substantial network of approximately 6000 bus stops. However, the geocoded information regarding all bus stops and the routes serving them is not publicly available on a single platform. To address this, geocoded 5579 bus stops on the ArcGIS 10.3 platform. We utilized secondary data sources, such as Google Maps, Moveit, and Chalo mobile applications, to associate each bus stop with the corresponding bus routes. In addition to its comprehensive bus network, Mumbai boasts a massively used suburban railway system comprising 53 stations.

It is important to note that Mumbai's metro system is being constructed. By 2026, approximately 337 km of metro with 130 stations will be operational in the Mumbai Metropolitan Region (MMR). Only Line 1 (Versova-Andheri-Ghatkopar Line), Line 2A (Dahisar-DN Nagar Line), and Line 7 (Andheri East-Dahisar East) are operational till March 2024. These three metro lines have a total length of 46.5 km and 42 stations (CTS, 2019). This study concentrates on the current situation in Mumbai: buses, metro, and suburban trains operations till March 2024. Having all transit infrastructure and demographic data from the same year is ideal; however, due to the COVID-19 pandemic, the planned 2021 census did not take place, and we use the most recent available population data from 2019 (CTS, 2019) to estimate the potential transit-dependent population (as explained in Section 4.2.1).

We tried to estimate the slum population at the TAZ level in Mumbai using a combination of GIS data, Google Earth satellite imagery, and population density-based approaches. The slum population data is available at the ward level; except for one ward, all wards have a slum population. The Slum Rehabilitation Authority (SRA), a statutory body under the Housing Department of the Government of Maharashtra, India, plays a pivotal role in slum redevelopment, seeking to improve the living conditions of these communities (SRA, 2025). We utilized the latest available SRA's slum cluster map (2015) to digitize the slum cluster and determine that approximately nine percent of Mumbai's area is covered by slums.

To estimate the slum population at the TAZ level, we first calculated the slum population density at the ward level, considering both the slum area and slum population in each ward. We applied this density to estimate the slum population in each TAZ, accounting for the slum area within the TAZ boundaries. We used the built-up area estimated for the year 2020 for Mumbai by Patil et al. (2025) to estimate the non-slum area (please refer to Patil et al. (2025) for further details on the built-up area estimation). Subtracting the slum area from the total built-up area allowed us to determine the non-slum built-up area, and the density-based approach is employed to estimate the non-slum population in each TAZ.

4. Methodology

Fig. 2 outlines the research methodology of the current study. The first primary step is estimating accessibility to the system through the Public Transport Accessibility Level (PTAL) method and evaluating the transit gap. The estimated PTAL at the TAZ level is used to analyse spatial equity among different population groups, which helps to estimate the transit gap. Furthermore, a structured decision-making framework is developed by integrating social vulnerability to propose policy recommendations.

4.1. Public transport accessibility level (PTAL)

PTAL is a highly origin-based transit accessibility measure that includes factors such as walking time from a point of interest to the service access point, waiting time at the transit stop, and the number of available transit routes at the stop. It helps understand how well a place is connected to public transport services; the higher the PTAL value, the better the area is connected to the public transit system. The method was originally developed for the London Borough of Hammersmith and Fulham in 1992 (Transport for London, 2010), current scores and methodology are available via WebCAT, an online tool launched by Transport for London in 2015 (WebCAT, 2025).

The PTAL method is suitable for urban cities with readily available transit services and is effective in data-scarce cities. Considering all these factors, we applied PTAL to estimate system accessibility in Mumbai, modifying specific parameters to reflect local requirements. Table 1 compares key parameter differences across cities. The steps of PTAL calculation for Mumbai are explained below:

Step 1: selection of point of interest (centroid of grid) and service access points (transit stations): To estimate the PTAL, the study area was segmented into 400 x 400 m grids, with the centroid of each grid considered as the point of interest (POI). Bus stops, suburban railway stations, and metro stations are considered as the service access points (SAP).

Step 2: calculating walk access time between POI and SAP: As per the guidelines of the Ministry of Housing and Urban Affairs (MoHUA, 2015), 400 m and 800 m are assumed to be the maximum walking distances for bus stops and railway stations, respectively. Given the poor availability of footpaths in Mumbai, often encroached by local shops and vendors, a walking pace of 60 m per minute is utilized

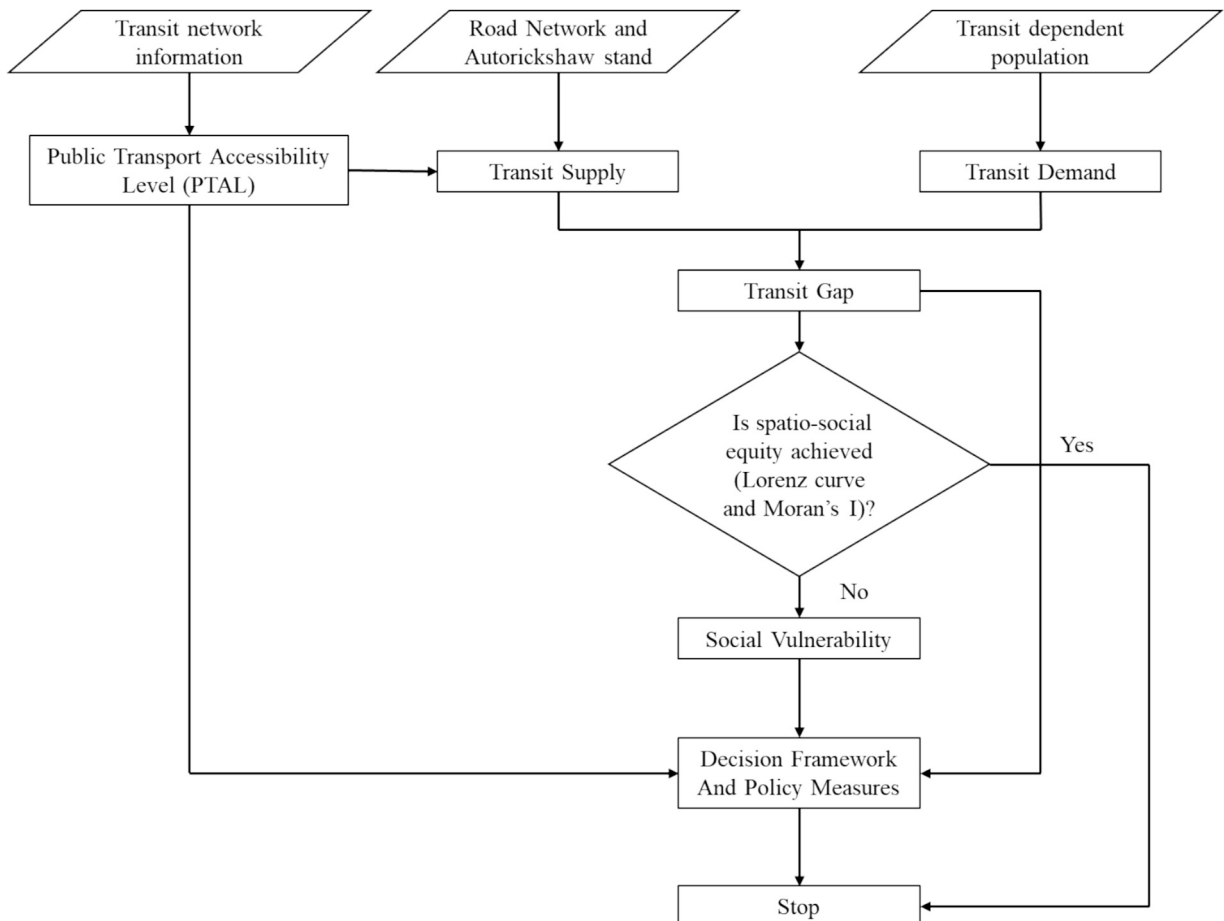


Fig. 2. Overview of the study methodology.

Table 1

Comparison of key parameters used in PTAL estimation across different cities and the rationale for Mumbai specific adaptations.

Parameter	London	Auckland	Melbourne	Ahmedabad	Surat	Lucknow	Singapore	Mumbai	Reason for Mumbai Adaptation
Source	Transport for London (2015)	Wu & Hine (2003)	Saghapour et al. (2016)	Shah & Adhvaryu (2016)	Adhvaryu et al. (2019)	Adhvaryu & Kumar (2021)	Yadav et al. (2024)	Current study	–
Peak hour (AM)	8:15 AM – 9:15 AM	7:00 AM – 10:00 AM	7:00 AM – 9:00 AM	09:30 AM – 10:30 AM	09:30 AM – 10:30 AM	9:00 AM – 10:00 AM	7:00 AM – 9:00 AM	8:00 AM – 11:00 AM	Based on the BEST and suburban train timetable
Walk Speed (km/h)	4.8	4.8	4.8	3.6	3.6	3.6	3.6	3.6	Poor availability of footpaths in Mumbai, often encroached by local shops and vendors
Reliability (K) for the bus system	2 min	2 min	Not included in the estimation	2.5 min	4 min	3 min	1.5 min	5 min	Based on the average delay survey
Reliability (K) for the railway system	0.75 min	0.75 min	Not included in the estimation	0.75 min	–	1 min	0.83 min	1.2 min	Based on the average delay survey
Maximum walk distance to the bus stop	640 m	640 m	800 m	993 m	900 m	900 m	400 m	400 m	As per the guidelines of the Ministry of Housing and Urban Affairs
Maximum walking distance to the railway station	960 m	960 m	1200 m	993 m	–	1200 m	800 m	800 m	As per the guidelines of the Ministry of Housing and Urban Affairs

for this study (Rastogi et al., 2011).

Step 3: Estimating scheduled waiting time (SWT) and average waiting time (AWT) for each route at each SAP: In PTAL, route data is accounted for only once per POI; if a bus or train serves multiple SAPs within a POI catchment area, the nearest SAP to the POI is selected. The average waiting time is calculated by adding the SWT with a reliability factor (K). The Comprehensive Mobility Plan (2016) (CMP-2016, 2016) and the Mumbai Suburban Rail Passenger Survey (2016) (Mumbai Railway Vikas Corporation Limited, 2016) highlight that the morning peak hour (8:00 AM to 11:00 AM) is more critical than the evening peak; thus, route and bus frequency data are used to estimate schedule waiting time (SWT). We conducted a five-day survey on weekdays during peak hours (8:00 AM to 11:00 AM). We estimated the delay in the bus's arrival at the bus stop from its scheduled time. We randomly selected approximately 10% of bus routes serving the city. Similarly, we surveyed the suburban railways on two major routes, i.e., the Western Railways and the Central Railways suburban routes. The observation indicates that the average potential delay (reliability factor) for buses in Mumbai is 5 min, while for trains, it is 1.2 min.

$$SWT_{ij} = 0.5 \times \left(\frac{60}{F_{ij}} \right) \quad (1)$$

$$AWT_{ij} = SWT_{ij} + K_j \quad (2)$$

Where SWT_{ij} is the schedule waiting time (min) at the nearest stop to the POI i for transit mode j , F_{ij} is the frequency of the mode j and at the nearest stop to the POI i , at the nearest stop to the POI i for public transit mode j , and K_j is the reliability factor.

Step 4: Estimation of total access time (TAT) and equivalent doorstep frequency (EDF) for each route at each SAP: The total access time (TAT) is calculated by adding the walking time to the SAP and the average waiting time at the SAP. The equivalent doorstep frequency (EDF) converts the TAT back into frequency units.

$$TAT_{ij} = WT_{ij} + AWT_{ij} \quad (3)$$

$$EDF_{ij} = 0.5 \times \left(\frac{60}{TAT_{ij}} \right) \quad (4)$$

Table 2

Descriptive statistics of variables used to develop the social vulnerability index (SVI) at the ward level.

Aspect	Normalised Indicator	Year	Minimum	Maximum	Mean	Standard deviation	Estimated weight
Demographic	Population density (persons per sq. km.)	2019	7603.3	86995.3	35896.3	18604.8	0.187
	% of child population (0–6 years)	2011	6.8	13.1	9.3	1.4	0.879
	% of the scheduled caste and tribe population	2011	1.6	18.9	7.5	3.4	0.449
	% of marginal workers	2017	1.7	3.2	2.4	0.4	0.470
	% of the working population	2017	34.5	52.5	41.1	4.6	0.671
Socio-economic	% of student population	2017	2.4	9.9	6.3	1.7	0.714
	% of households not having a car/jeep/van	2011	70.7	95.2	86.9	6.9	0.483
	% of households not having a television	2011	7.7	43.5	15.8	7.0	0.723
	% of households not availing banking services	2011	5.8	42.5	14.2	7.9	0.820
	% of the population in the slum	2011	10.0	84.9	44.4	22.9	0.665
Housing and Hygiene Condition	Person illiteracy rate (%)	2011	6.8	16.6	10.2	2.0	0.631
	% of households not connected to treated water	2011	0.9	4.8	2.2	0.9	0.682
	% of households not connected with electricity as the primary source of lighting	2011	0.3	35.7	3.9	6.7	0.589
	% of households not having a toilet facility within the premises	2011	6.4	67.9	39.2	16.0	0.928
	% of household wastewater outlet not connected to closed drainage	2011	1.3	47.9	16.8	12.8	0.891
Education, Health Status, and Infrastructure	% of malnourished students in municipal schools	2017	4.0	51.0	20.0	13.2	0.679
	% of household where no member has health insurance	2011	39.0	80.0	72.4	8.9	0.165
	Total death	2017	1019.0	7203.0	3885.6	1542.4	0.248
	Number of healthcare centres per 100 K population	2017	0.8	5.7	2.2	1.2	0.310
	Number of government schools per 10 K students	2017	12.2	44.1	23.5	8.4	0.692
	Road Density (km. per sq. km.)	2020	1.7	31.5	9.3	6.9	0.512

Where TAT_{ij} is the total access time (min) of public transit mode j at the nearest stop to the POI i , WT_{ij} is the walk time (min) from the POI i to the nearest stop of public transport mode j and EDF_{ij} is the equivalent frequency for public transport mode j at the nearest stop to the POI i .

Step 5: Estimate Access Index and PTAL: Finally, the EDFs of all routes passing through SAPs within the walking distance catchment area are summed to determine the PTAL value for a POI. One weight is assigned to the highest EDF per mode, and half is given to all others. The equations used for each step are mentioned below.

$$AI_{ij} = EDF_{ij_{max}} + \left(0.5 \times \sum_{j \neq j_{max}} EDF_{ij} \right) \quad (5)$$

$$PTAL_i = \sum_j AI_{ij} \quad (6)$$

Where AI_{ij} is the access index of public transit mode j at the nearest stop to the POI i , and $PTAL_i$ is the Public Transport Accessibility Level for a POI i served by all modes. Here modes considered are bus, suburban train, and metro.

Area-weighted PTAL values are calculated after excluding reserved areas. The estimated accessibility levels are classified into six groups for mapping, consistent with the studies by [Yadav et al. \(2024\)](#) and [Saghapour et al. \(2016\)](#) (i) Very poor (0–5) (ii) Poor (5–10) (iii) Moderate (10–15) (iv) Good (15–20) (v) Very Good (20–25) and Excellent (>25). To test the robustness of the adopted PTAL classification, we varied the threshold values by $\pm 10\%$. The analysis showed only minor shifts in category proportions. At the same time, the overall distribution pattern remained consistent, with higher accessibility areas continuing to contain a larger share of the population regardless of threshold adjustments, confirming the reliability of the classification.

4.2. Transit gap

4.2.1. Transit demand

In this study, the potential transit dependent population is estimated based on individuals without access to personal vehicles and those likely to rely on public transport due to age (e.g., minors and the elderly). The assumption is that individuals without private vehicles depend on public transport for their commute. In India, the minimum age for obtaining a driving license is 18, and a medical certificate is required for license renewal after age 40. License renewal is required every five years after the age of 60 ([Motor Vehicles Department, 2025](#)). Therefore, it is assumed that, due to these constraints, the elderly and minors rely on public transport for their

daily commute. The U.S. Department of Transportation originally developed this approach, and it has been widely adopted in various transit desert studies (Aman & Smith-Colin, 2020; Cai et al., 2020; Chen & Jiao, 2022; Jiao, 2017; Jiao & Dillivan, 2013; Lee et al., 2021; Yadav et al., 2024). A similar approach is used in this study and is tailored to suit Mumbai's data availability.

Due to the COVID-19 pandemic, the regular census was not conducted in 2021. As a result, we relied on the 2019 projected population data provided in the Comprehensive Transport Study for Mumbai (CTS, 2019), which reported an average occupancy of 2.3 persons per car and 1.4 persons per motorbike for work trips in Mumbai. For their daily commute to school, high school or secondary school students in Mumbai also frequently use public transport.

To estimate the potential transit dependent population by TAZ, the combined populations of minors (aged 15 to 20 years), adults of working age (aged 21 to 60 years), and the elderly population (aged 60 and over) are considered. To refine this estimate, the number of motorbikes and cars in each TAZ was subtracted and adjusted using the respective average occupancy rates. For this study, we define the "Transit Dependent Population" (TDP) to represent the potential transit dependent population in each TAZ.

$$TDP_{\text{taaz}} = P_{\text{taaz}} - (C_{\text{taaz}} \times O_c) - (TW_{\text{taaz}} \times O_{\text{tw}}) \quad (7)$$

Where, P_{taaz} represents the population aged fifteen years and above, C_{taaz} and TW_{taaz} denote the number of cars and two-wheelers in a given TAZ, respectively, and O_c and O_{tw} correspond to the average passenger occupancy per car and per two-wheeler.

To avoid underestimating Mumbai's potential transit-dependent population (6.6 million). We validated our estimate using publicly available data, ensuring no double-counting. This validation included the sum of public transit commuters to work (3.5 million) (CTS, 2019), minors (aged 15 to 20 years) (1.2 million) (Praja Organisation, 2017a), the elderly (those aged 60 and above who are still working) (0.82 million) (Raju et al., 2016), and the unemployed population (those who are not working but actively seeking employment) (0.79 million) (CTS, 2019). The difference between our estimated values and the validation data is less than 5%.

After estimating the transit-dependent population, the values are divided by the TAZ area to calculate population density and nullify the size effect. The z-scores for each TAZ are then computed. The z-score, or standard score, indicates how a particular value deviates from the mean of a dataset (Lee et al., 2021).

4.2.2. Transit supply

This study uses PTAL while calculating the transit supply to address the interconnection challenges of transit infrastructure and service. PTAL incorporates factors such as walkability to public transport stops, selection of feasible routes, and transit service frequency, offering a more wholesome assessment of transit supply.

Factors such as sidewalk density, cycle stand availability, and intersection density contribute to improving access to the transit system, but do not directly indicate the level of transit supply (Aman & Smith-Colin, 2020). In Mumbai, where non-motorized transport infrastructures such as sidewalks, cycle tracks, and bike-share facilities are generally inadequate or absent, auto-rickshaws and taxis play an important role in bridging the first and last mile to formal transit services, including suburban rail and metro. These modes operate from officially designated stands located near suburban railway and metro stations. During peak hours, they serve as structured feeder services, supporting seamless commuter transfers. Recent studies by Baro et al. (2025) and Abhijna et al. (2025) highlight this role, reporting that nearly 30% of suburban and metro rail commuters rely on auto-rickshaws or taxis to reach stations.

This study estimates transit supply by combining three indicators: PTAL scores, road density, and auto-taxi stand density. PTAL captures formal public transport availability based on bus and train service. Due to the lack of footpath data, road density (excluding expressways) is used as a proxy for pedestrian infrastructure. Furthermore, auto-taxi stand density is incorporated to represent the role

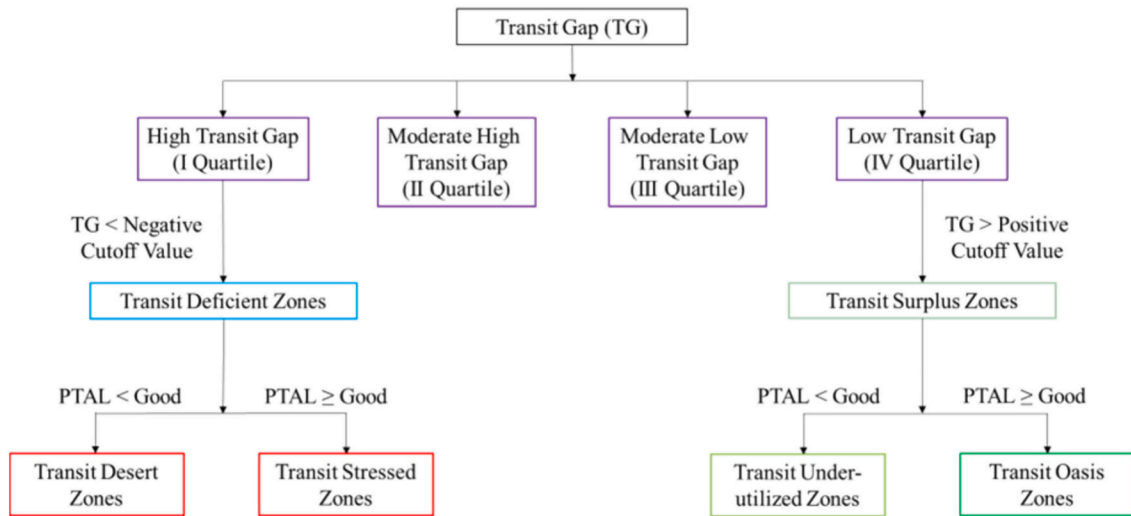


Fig. 3. Framework for defining transit desert and transit oasis adopted from Yadav et al. (2024).

of informal but organized last-mile services in enhancing accessibility to the transit system.

$$Z_{supply} = (Z_{PTAL} + Z_{Road\ density} + Z_{Auto-Taxi\ stand\ density})/3 \quad (8)$$

Where, Z_{PTAL} is Z-score of the PTAL value for a TAZ, $Z_{Road\ density}$ is Z-score of road density for a TAZ, $Z_{Auto-Taxi\ stand\ density}$ is Z-score of auto-taxi stand density for a TAZ.

4.2.3. Transit gap and transit desert

The transit gap is determined by subtracting transit demand from transit supply. A positive value indicates supply exceeding demand, while a negative value signifies greater demand than supply. The transit gap for each TAZ is calculated using the equation provided below:

$$Transit\ Gap_{taaz} = Transit\ Supply_{taaz} - Transit\ Demand_{taaz} \quad (9)$$

Following previous literature that supports the use of quartile methods for classifying maps (Aman & Smith-Colin, 2020; Milic et al., 2019; Roy et al., 2023), the scores of transit supply, transit demand, and transit gap are categorised into quartiles. Each quartile represents one-fourth of the subzones. This classification approach helps analyse the percentage of the population residing in subzones experiencing the most significant transit gaps. To test the robustness of the transit gap thresholds, we varied the quartile cut-offs by $\pm 10\%$. The results showed only minor shifts in category proportions ($\pm 4.4\%$ for high gap and within $\pm 2.3\%$ for others), while the overall distribution pattern remained consistent, confirming the reliability of the classification.

This study adopts the framework developed by Yadav et al. (2024), which uses a cross-classification of transit gap and PTAL to redefine transit deserts and transit oases, see Fig. 3. This framework is applied to identify TAZs functioning as transit deserts. The classification framework divides zones into four categories by combining the transit gap z-score distribution (positive and negative thresholds) with PTAL levels (Good and Below Good), allowing a more nuanced understanding of supply–demand mismatches. This approach ensures that areas are not labelled transit deserts solely based on the negative transit gap alone, but on the interaction with PTAL. A TAZ is identified as a ‘transit desert’ if its transit gap is below the negative threshold and its PTAL is below the Good Level.

This study uses threshold values of -1.0 and 1.0 , aligning with Mumbai’s transit patterns similar to other Asian (Cai et al., 2020; Yadav et al., 2024) and European cities (Chen & Jiao, 2022). TAZs with a transit gap below the negative threshold and a PTAL at or above the Good Level are labelled ‘transit stressed zones’. This method distinguishes between areas with high transit gaps due to inadequate service (transit desert) and those with adequate service but high transit dependency (transit stressed area). Conversely, a ‘transit oasis’ is where the transit gap surpasses the positive threshold and the PTAL meets or exceeds the Good Level. Areas experiencing a transit gap above the positive threshold but a PTAL below the Good Level are termed ‘transit underutilized zones,’ needing more transit-dependent populations to utilize the available transit effectively. This classification underscores the importance of transit supply, recognizing that transport and city planners have more authority over transit supply than demand.

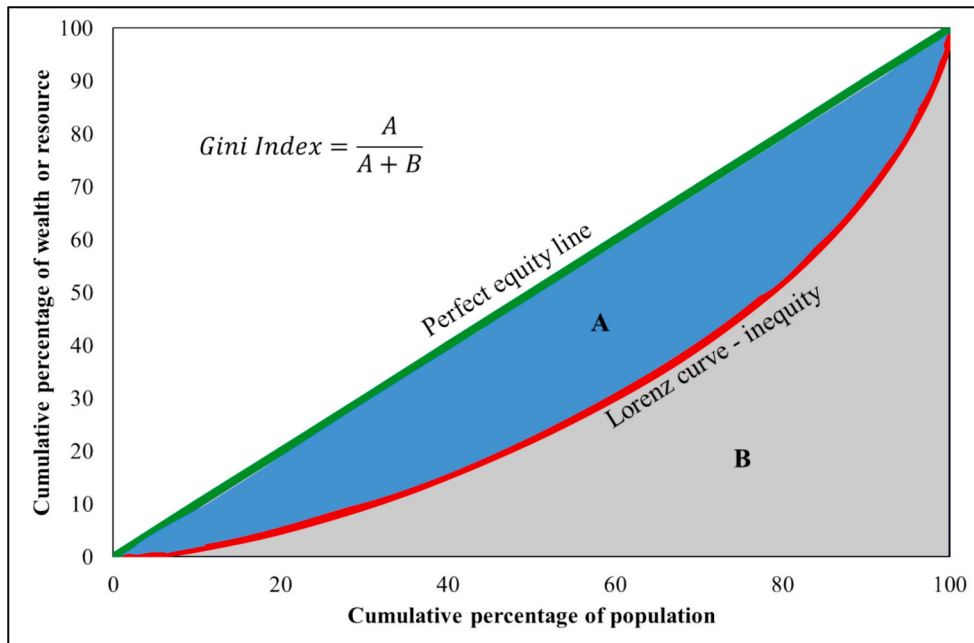


Fig. 4. Lorenz curve and Gini Index.

4.3. Equity assessment for transit system accessibility

4.3.1. Lorenz curve

The Lorenz curve illustrates the distribution of a resource or attribute, such as income or wealth, across a population, with the line of perfect equity, see Fig. 4. In this study, the Lorenz curve represents the cumulative share of the population against the cumulative share of PTAL available to them.

The greater the deviation of the Lorenz curve from the line of perfect equity, the higher the level of inequity. The Gini index is often used to quantify the inequity that the Lorenz curve depicts. The Gini index is the area between the perfect equity line (green line in Fig. 4) and the Lorenz curve (red curve in Fig. 4) represented by the blue area in Fig. 4, divided by the total area under the line of perfect equity (the sum of the blue and the gray regions in Fig. 4) (Sharma & Patil, 2021; Tahmasbi et al., 2019). Gini index can be estimated using the trapezoidal method based on interval differences, as in Equation (10) (Roy et al., 2024b; Tahmasbi et al., 2019).

$$G = 1 - \sum_{i=1}^{n-1} (X_{i+1} - X_i)(Y_{i+1} + Y_i) \quad (10)$$

Where G is the Gini index, X_i is the cumulative proportion of the population over n traffic analysis zone and Y_i is the cumulative proportion of PTAL.

The Gini index ranges from 0 to 1, where values near 0 indicate high equity with fair wealth distribution and those near 1 indicate low equity and significant wealth disparity across the population (Tahmasbi et al., 2019). In the context of transit system accessibility, a high Gini index would suggest substantial disparities in access to transit services among a population group. For public transit accessibility, the Gini index is commonly interpreted using the following thresholds: values below 0.20 indicate low inequity, values between 0.20 and 0.50 represent moderate inequity, and values above 0.50 reflect high inequity (Jang et al., 2017; Ni et al., 2024). These cutoffs are relatively insensitive to outliers at either end of the distribution (Jang et al., 2017).

4.3.2. Bivariate local indicators of spatial Association (BiLISA)

Although the Lorenz curve and Gini index effectively illustrate the overall degree of inequity, they have limitations in spatial analysis. To fill this gap, we use Bivariate Global Moran's I , a spatial autocorrelation statistic measuring the degree of correlation between two different spatial variables over a geographical area. A positive value of Bivariate Global Moran's I indicates that areas with high values of one variable are often situated near areas with high values of the other variable, and similarly, low values tend to cluster together. In contrast, a negative value suggests an inverse relationship, where high values of one variable are generally found near low values of the other. BiLISA, on the other hand, provides a local measure of spatial association, identifying specific locations where the correlation between the two variables is particularly strong or weak. This helps pinpoint areas of significant spatial clustering or dispersion that might be masked in the global statistics. The formula for global bivariate Moran's I (I_{AP}) and BiLISA (I_{AP}^L) is as follows (Jin et al., 2022):

$$I_{AP} = \left(\frac{n}{\sum_i \sum_j w_{ij}} \right) \left(\frac{\sum_i \sum_j w_{ij} (A_i - \bar{A})(P_j - \bar{P})}{\sqrt{\sum_i (A_i - \bar{A})^2 \sum_j (P_j - \bar{P})^2}} \right) \quad (11)$$

$$I_{AP}^L = \frac{n}{\sum_i (x_i - \bar{x})^2} (A_i - \bar{A}) \sum_j w_{ij} (P_j - \bar{P}) \quad (12)$$

Where n represents the number of spatial units, such as TAZs or wards. A_i and P_j are the values of the transit gap and population at locations i and j , respectively. $\bar{A}$ and $\bar{P}$ are the mean values of the transit gap and population variables. The spatial weight w_{ij} indicates the spatial relationship between the two locations. In this study, the weight matrix is generated assuming queen contiguity, where w_{ij} is 1 if locations i and j share a common edge or vertex, and 0 otherwise.

4.4. Social vulnerability index

The Social Vulnerability Index (SVI) is a comprehensive tool designed to quantify social vulnerabilities and identify populations at risk. Similar to the themes provided by the Centers for Disease Control and Prevention (CDC) in the United States for developing an SVI for disaster management, we grouped our 21 variables into four significant attributes: demographic, socioeconomic, education, health and infrastructure, and housing and hygiene. The variables and their descriptive analysis included in these groups are presented in Table 2. All the data at the ward level is collected from the Census of India (Census of India, 2011) and the Praja Foundation reports (Praja Organisation, 2017a, 2017b) on Mumbai.

A critical step in constructing the SVI is assigning appropriate weights to the indicators. Assuming equal contribution from all indicators may oversimplify the complex interplay of socio-economic and infrastructural factors. To address this, Principal Component Analysis (PCA) is often used in index construction, as it assigns weights based on the variance. It assigns higher weight to the indicators exhibiting greater variance across spatial units (i.e., wards). This is a reasonable assumption because if an indicator has similar values across all wards, whether good or bad, it shows little variation (i.e., a low standard deviation). As a result, it doesn't help much in distinguishing between areas, so it receives a lower weight in the SVI calculation (Maiti et al., 2017; Sharma & Patil, 2021).

PCA is a multivariate statistical technique that reduces dataset dimensionality by transforming it based on linear relationships among variables, helping identify latent information within the data (Roy et al., 2022). Indicators used in PCA must be scale independent. Indicators contributing to exposure and sensitivity are positively associated with vulnerability, while those enhancing adaptive capacity are inversely related. Out of 21 variables, 17 contribute to exposure and sensitivity capacity, and four variables (percentage of working population, number of healthcare centres per 100,000 population, number of government schools per 10,000 students, and road density) contribute to adaptive capacity. Variables are normalized using Equations (13) and (14) (Majumder et al., 2023). Equation (13) is used for variables expressing exposure and sensitivity capacity, i.e., those positively influencing social vulnerability. In contrast, Equation (14) is used for variables expressing adaptive capacity, i.e., those that negatively affect social vulnerability.

$$X_{ij}^{Pos} = \frac{Y_{ij} - \text{Min}(Y_{ij})}{\text{Max}(Y_{ij}) - \text{Min}(Y_{ij})} \text{ (for } Y_{ij} \text{ being a positive indicator)} \quad (13)$$

$$X_{ij}^{Neg} = \frac{\text{Max}(Y_{ij}) - Y_{ij}}{\text{Max}(Y_{ij}) - \text{Min}(Y_{ij})} \text{ (for } Y_{ij} \text{ being a negative indicator)} \quad (14)$$

Where, X_{ij}^{Pos} or X_{ij}^{Neg} is a normalised value of j^{th} variable for the ward i . Y_{ij} , $\text{Min}(Y_{ij})$ and $\text{Max}(Y_{ij})$ are the observed, minimum, and maximum values of the j^{th} variable for the ward i .

As established in previous studies (Maiti et al., 2017; Roy et al., 2023, 2024a; Sharma & Patil, 2021), the loadings of the first principal component are commonly used as weights, as this alone captures the maximum variance in the dataset. Following this, we derive indicator weights from the first component, which explains 40.7% of the total variance. Estimation of social vulnerability index for a ward (SVI_i) can be performed using Equation (15).

$$SVI_i = \sum_{j=1}^J W_j X_{ij} \quad (15)$$

Where, W_j is the weight corresponding to the normalised variable X_{ij} , derived from the first principal component.

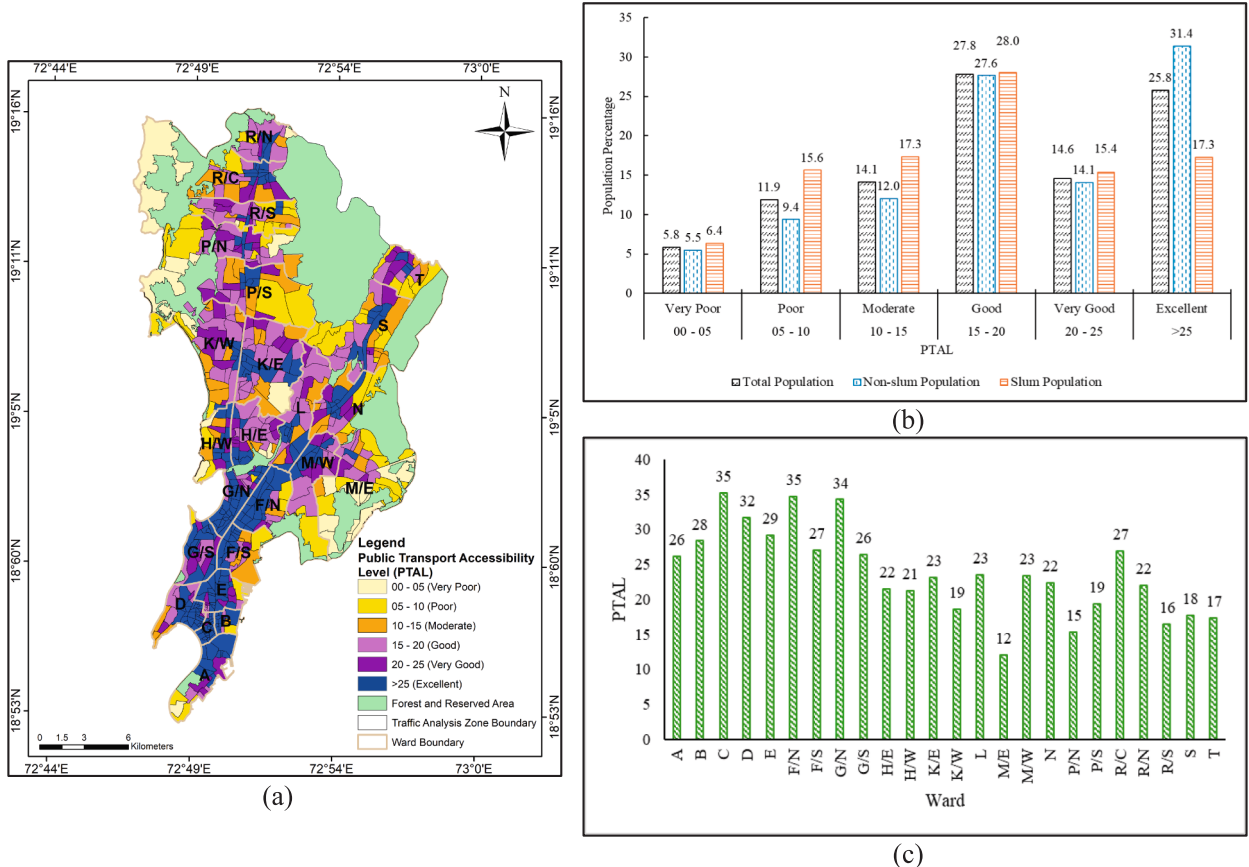


Fig. 5. (a) Spatial distribution of PTAL, (b) population staying in different PTAL ranges, and (c) average PTAL value for the wards in Mumbai.

5. Results

5.1. Mumbai PTAL mapping

The PTAL analysis results demonstrate varying levels of public transport accessibility across slum and non-slum populations. Fig. 5 illustrates (a) the spatial distribution of PTAL, (b) the population residing in different PTAL ranges, and (c) average PTAL values for wards in Mumbai. Among Mumbai's 640 TAZs, 22 zones, accounting for 5.8% of the total population, have very poor accessibility. Additionally, 57 zones (15.6% of the slum population and 9.4% of the non-slum population) are classified as poor, and 68 zones (17.3% of the slum population and 12.0% of the non-slum population) as moderate. In other words, 147 TAZs accounting for 31.8% of the total population do not have a satisfactory PTAL. Only 17.3% of the slum population has excellent transit accessibility, compared to 31.4% of the non-slum population.

Fig. 5 (c) shows the ward-level PTAL values obtained by taking the population-weighted average (Sharma & Patil, 2021) of PTAL values of TAZs within that ward. It is noted that wards in the south (A, B, C, D, E, G/S, G/N, F/S, and F/N), where 27.61% of the non-slum population and only 15.11% of the slum population reside, have significantly higher accessibility levels compared to the rest of the city. The bottom five wards regarding PTAL values are M/E, P/N, S, T, and R/S. Wards M/E, S, and T are located on the city's eastern side, while P/N and R/S are situated in the northern part. Together, these five wards are home to 32.57% of the slum population and 26.87% of the non-slum population.

The PTAL analysis reveals notable spatial inequity in transit supply across Mumbai. These results also highlight significant horizontal inequity in system accessibility within the same population group and vertical inequity between slum and non-slum populations. The southern and western parts of the city generally enjoy better public transport access than the northern and eastern regions. Addressing these inequalities is crucial for creating a more equitable urban transit system.

5.2. Spatial distribution of transit Demand, transit Supply, and transit gap

Figs. 6(a) and 6(b) illustrate the spatial distribution of z-scores for transit demand and supply, respectively. The southern part of the city shows the lowest transit demand, primarily due to the predominance of commercial establishments and lower residential density. This lower demand may be attributed to higher housing costs and favourable financial conditions in these TAZs, which increase private vehicle ownership and reduce dependence on public transport (Gandhi et al., 2014). Conversely, high transit demand TAZs (zones lying in the fourth quartile) are clustered mainly in the eastern and western wards. These high transit demand zones encompass 40.5%

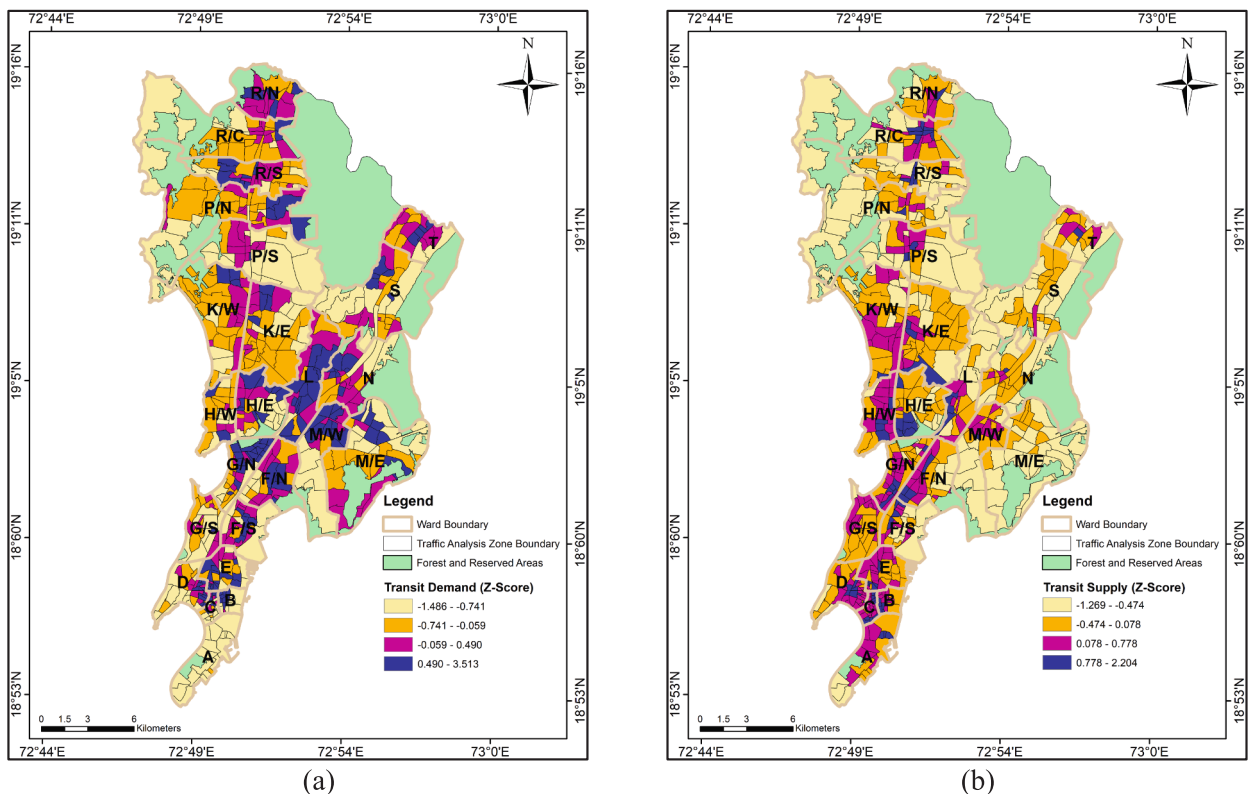


Fig. 6. Spatial distribution of (a) transit demand (z score), (b) transit supply (z score).

of the total slum areas, housing 44.4% of the slum population. For the non-slum population, 21.3% reside in the highest demand areas (IV Quartile).

In the lowest quartile range (first quartile) of transit supply, primarily visible in the northern and eastern part of the city, 33.9% of the population resides. In contrast, TAZs in the southern and western parts perform better in supply indicators. Notably, 41.1% of the slum population resides in the lowest quartile of supply, covering 49.3% of slum areas. Conversely, 13% of the slum population resides in the highest quartile (fourth quartile) of supply, covering 6.8% of the slum areas. For the non-slum population, 17% reside in the highest supply levels.

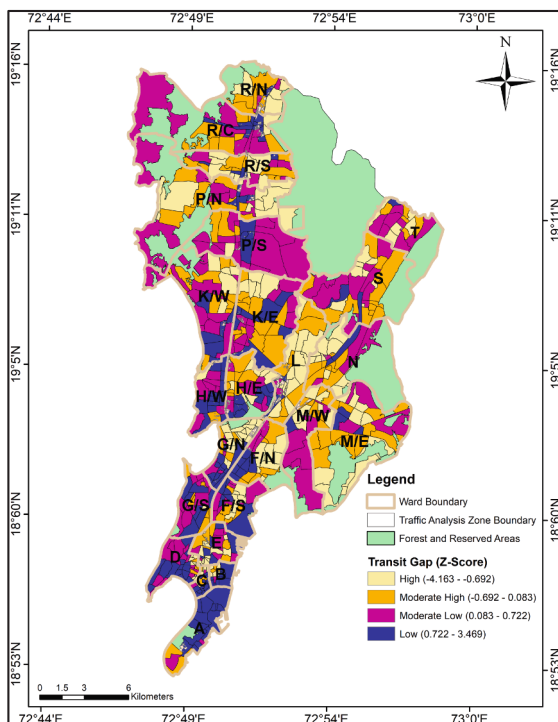
Figs. 7(a) and 7(b) depict the spatial distribution of the transit gap and population in different transit gap ranges, respectively. Notably, 39.1% of the total population resides in high transit gap zones. Zones with high transit gaps are distributed across the city, forming clusters mainly in Mumbai's E, G/N, F/N, L, N, M/E, K/E, and P/N wards. Over half of the slum population resides in zones classified as high transit gap, while around one-third of the non-slum population also falls under the same criteria. Additionally, 38.7% of the non-slum population lives in areas where supply exceeds demand, i.e., moderate-low and low transit gap regions. In comparison, this number for slums is only 18.9%, indicating better system accessibility in non-slum areas than in slum areas.

The transit gap analysis highlights significant inequity in accessibility to transit stations between slum and non-slum populations in Mumbai. Notably, some slum areas have better access to transit systems than other slum clusters, indicating variability within the slum population. Similarly, inequity exists within the non-slum population, where certain areas benefit from superior transit accessibility while others fall short.

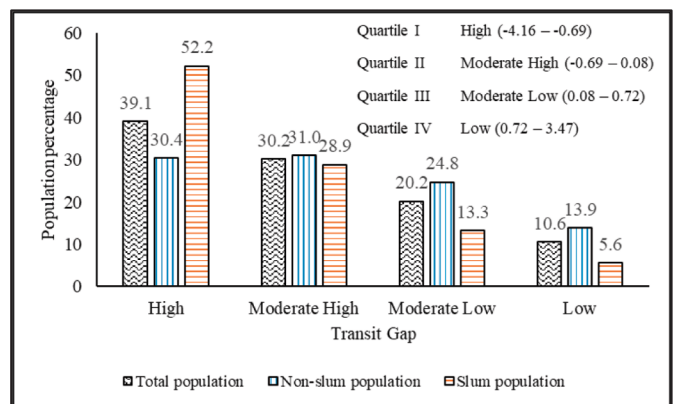
6. Socio-spatial equity assessment

The Lorenz curve and Gini index analysis for Mumbai reveal critical insights into the degree of inequities in transit supply both within the same population groups and among different population groups (see Fig. 8). The analysis is carried out with reference to population type (slum population and non-slum population) and employment type (office job and industry job). We examined the employed population based on their residence location (residential workforce) and work location (workplace employment).

For the total population (Fig. 8(a)), the inequity in PTAL is moderate ($G = 0.48$). The higher Gini index of the slum population reflects that, within the slum population, inequity in transit access is higher, likely due to the poor transit distribution around slum areas. The lower Gini index value for the non-slum population indicates that the transit system is better distributed within the non-slum population than the slum population. However, the Gini value also suggests a moderate level ($G = 0.41$) of inequity within the non-slum population, implying that a significant portion of this group still lacks adequate access to public transport. This may be due to several factors: economically constrained individuals in non-slum areas may reside farther from city centres or major transit corridors,



(a)



(b)

Fig. 7. (a) Spatial distribution of transit gap and (b) population staying in different transit gap ranges.

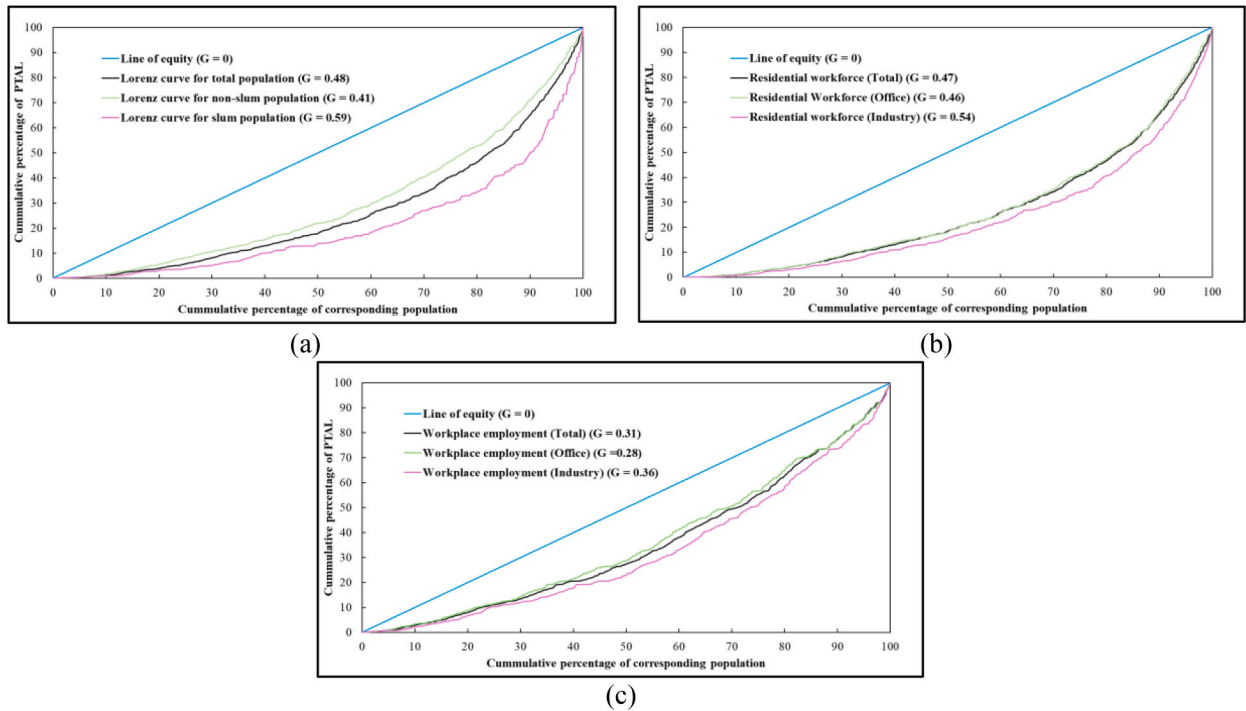


Fig. 8. Lorenz curve – (a) Population group, (b) Residential workforce, and (c) Workplace employment.

where housing is more affordable, but transit services are less frequent. Additionally, some non-slum areas are located near forest zones or reserved lands (see PTAL Map, Fig. 5), where transit provision is limited due to environmental and planning restrictions. These geographic and socioeconomic constraints contribute to the moderate inequity observed within the non-slum group, despite its better infrastructure than slum areas.

The residential workforce, representing employees living in the same TAZ (Fig. 8(b)), shows a moderate degree ($G = 0.47$) of inequity. Challenges within the industrial workforce are more than those of office workers, as reflected by the Lorenz curve. Workplace employment measures the availability of job opportunities in a TAZ (workplace location) and shows a moderate degree of inequity ($G = 0.31$) as seen in Fig. 8(c). This result suggests that workplace locations are generally better served by transit, as major employment hubs, primarily in the southern and western parts of the city, exhibit higher PTAL values than the more dispersed residential distribution.

There is a moderate level of inequity within industrial jobs, while within office jobs, inequity is minimal among all classes. These results indicate that office workers are more likely to reside in TAZs with low PTAL and work in TAZs with better-managed transit supply. In contrast, industrial workers face worse conditions in both residential and workplace TAZs, highlighting significant inequity in transit access. This may highlight a job-housing mismatch, particularly for the urban poor who are more likely to live on the periphery and remain highly dependent on public transport for commuting (Gandhi et al., 2014).

Although the Lorenz curve and Gini index effectively depict the overall degree of inequity, they have limitations in spatial analysis. To fill this gap, we use Bivariate Global Moran's I . Table 3 presents bivariate Moran's I values, significant at p -value less than 0.001. To remove the size variation of different TAZs, we divided the population and employment counts by TAZ area to convert into population and employment densities.

Table 3
Bivariate global Moran's I value for various population groups and employment categories.

Class (Density)	Bivariate Global Moran's I
Total Population	-0.215
Non-slum Population	0.015
Slum Population	-0.262
Residential workforce (Total)	-0.214
Residential workforce (Industry)	-0.256
Residential workforce (Office)	-0.073
Workplace employment (Total)	0.184
Workplace employment (Office)	0.205
Workplace employment (Industry)	0.073

The Bivariate Local Indicators of Spatial Association (BiLISA) results, shown in Fig. 9, reveal important spatial clustering patterns between transit gap and population density subgroups. These spatial patterns significance level maps are illustrated in Fig. A1-1 (a, b, and c) in Appendix 1. The Bivariate Global Moran's I statistic shows a negative spatial correlation with the transit gap for the total population density. This suggests that TAZs with low transit gaps, where transit supply meets demand, do not cluster with high population densities. Similarly, the slum population shows a relatively stronger negative correlation, indicating that slum areas with high population densities are concentrated in TAZs with high transit gaps. In contrast, the non-slum population shows a minor positive spatial correlation with the transit gap, suggesting a random distribution across areas with varying transit gaps. For the total population, 57 TAZs are classified as high-high zones, primarily located in the southern and western parts of Mumbai, while 71 TAZs are classified as low-low zones, distributed across the city. For the non-slum population, two clusters appear in the southern part of Mumbai, where the transit gap and population density are high-high. However, the PTAL map (refer to Fig. 5 (a)) indicates excellent PTAL levels in these areas, showing high demand and supply conditions. These TAZs in the southern part of Mumbai are surrounded by the low-high zones, which help by providing additional transit support. Conversely, for the slum population, many high transit gaps and high slum population density clusters arise, primarily in the eastern and northern parts of Mumbai, where the PTAL is up to a good range. These spatial patterns indicate that transit inadequacy is not randomly distributed but spatially concentrated in specific neighbourhoods.

Fig. 10 illustrates the local spatial correlation between transit gaps and the residential workforce (refer to Fig. A1-1 (d, e, and f maps) in Appendix 1 for the significance level. The residential workforce density shows a negative spatial correlation, similar to the total population. The BiLISA map shows that 50 TAZs fall into the high-high category, primarily forming three clusters located in wards L, G/N, and C. These results indicate that areas with high residential density surround zones with a high transit gap. However, the high transit gap in L ward arises due to a lack of public transport access, unlike in G/N and C wards. Industrial workers exhibit a stronger negative correlation, suggesting that TAZs, where industrial workers live, are clustered in regions with higher transit gaps. Office residential workers don't display any significant correlation, indicating a similar pattern shown by the non-slum population. The office-going workforce shows high-high clusters in some wards of southern Mumbai, where transit supply is generally good. In contrast, the industrial workforce shows high-high clusters in TAZs of the eastern ward of the city. This pattern aligns with the spatial distribution of slum populations, inferring that many slum residents work in industrial TAZs.

Workplace employment (employment locations) shows positive spatial correlations, indicating that areas with more job opportunities tend to have lower transit gaps, see Fig. 11 (refer to Fig. A1-1 (g, h, and i maps) in Appendix 1 for significance level). Out of 22 high-high TAZs for total workplace employment, 19 are clustered together, indicating that employment distribution is significantly higher in southern Mumbai compared to the rest of the city. Office employment locations have a higher positive correlation than industrial locations, indicating that regions with a concentration of office jobs are likelier to experience low transit gaps. For work locations, high transit gaps and high employment density zones arise primarily in the southern part of Mumbai wards for both kinds of employment. This occurs because these wards serve as the central business district for the entire Mumbai Metropolitan Region (MMR). Additionally, these high-high clusters for work locations are surrounded by low transit gaps and high employment opportunity TAZs, which can help balance the excess demand of high-high TAZs.

7. Vulnerable population and policy measures

In developing cities, transport planners must identify areas needing closer attention to enhance system accessibility and reduce

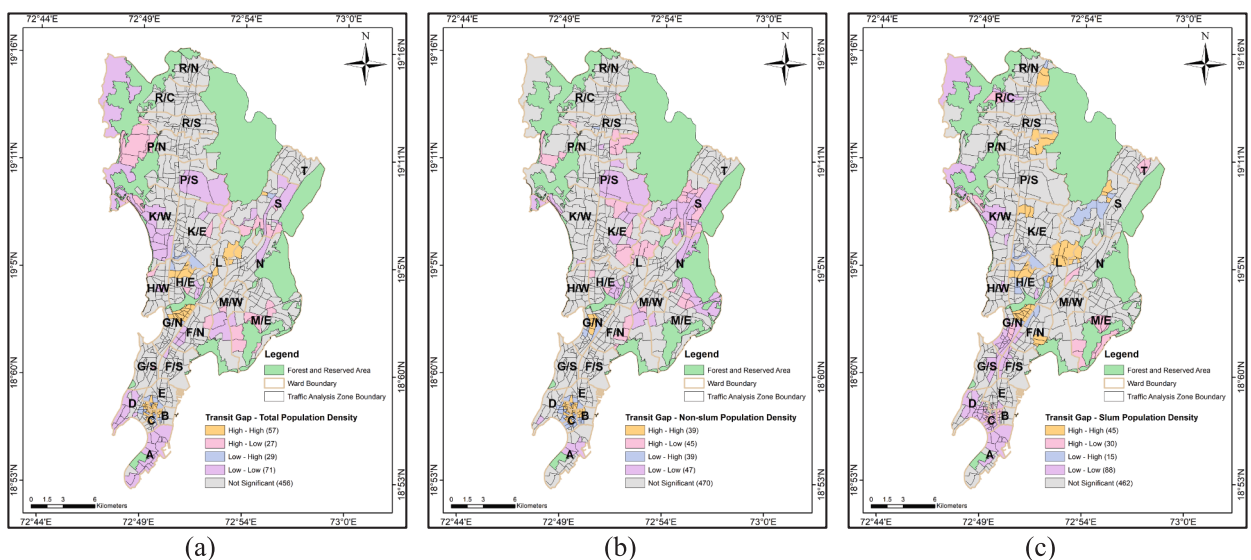


Fig. 9. BiLISA map for transit gap and (a) total population density, (b) non-slum population density and (c) slum population density.

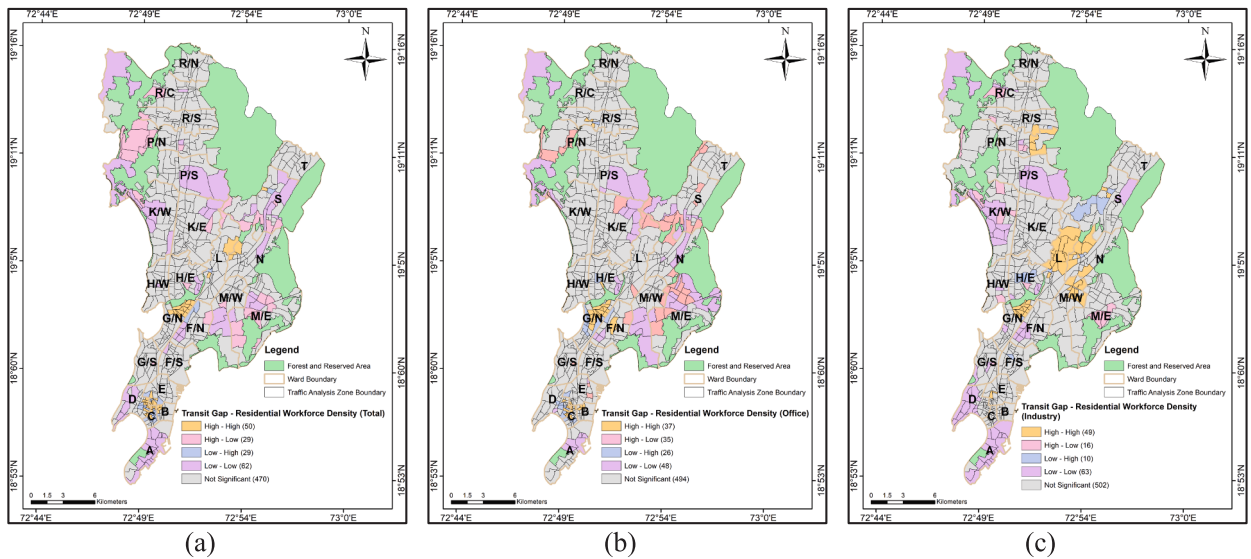


Fig. 10. BiLISA map for transit gap and (a) residential workforce density (Total), (b) residential workforce density (Office), and (c) residential workforce density (Industry).

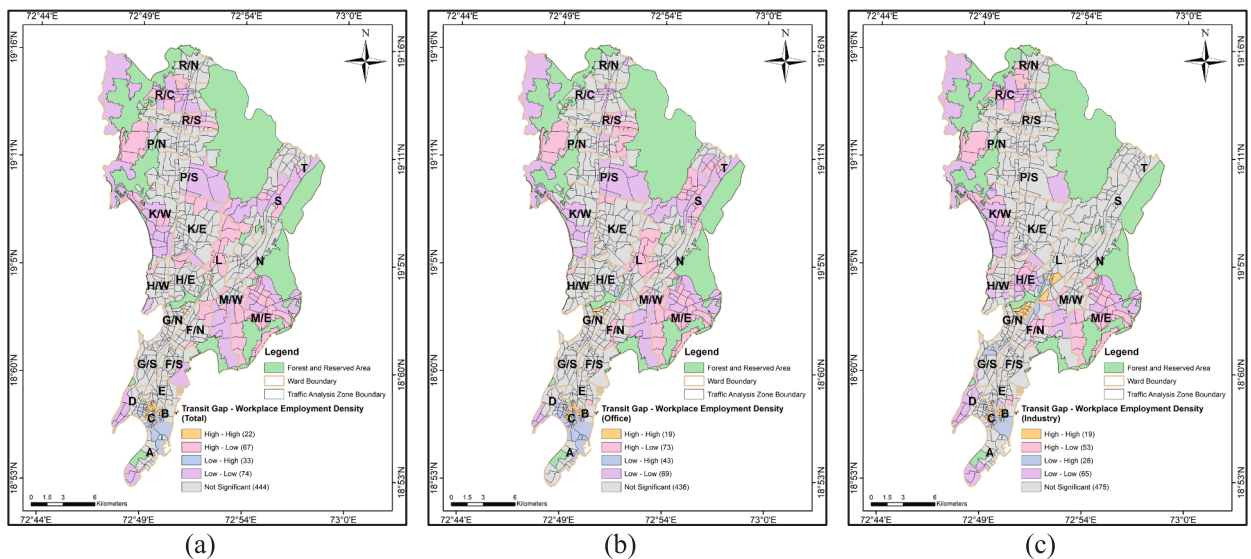


Fig. 11. BiLISA map for transit gap and (a) Workplace employment density (Total), (b) Workplace employment density (Office), and (c) Workplace employment density (Industry).

inequities in transit access. Simply considering slum and non-slum populations may not fully capture the city's dynamics; analysing vulnerable populations with transit accessibility measures offers a more comprehensive solution for identifying underserved areas.

7.1. Social vulnerability index

Fig. 12 shows the status of Mumbai wards in terms of social vulnerability. All wards in the city's eastern part, except ward T, are classified as highly vulnerable. Notably, several wards, such as K/W and K/E, despite having large slum clusters, performed better in terms of other indicators, which helps them resist vulnerabilities; hence, these wards lie in the low vulnerability category. Regarding population, 44.80% reside in low-vulnerability wards, while 55.20% live in high-vulnerability wards. For non-slum and slum populations, 52.12% of the non-slum population and 33.71% of the slum population live in low-vulnerability wards. Conversely, 47.88% of the non-slum population and 66.29% of the slum population reside in high-vulnerability wards.

The results from BiLISA (Figs. 9-11) and the SVI analysis (Fig. 12) indicate that several high-high clusters, based on total

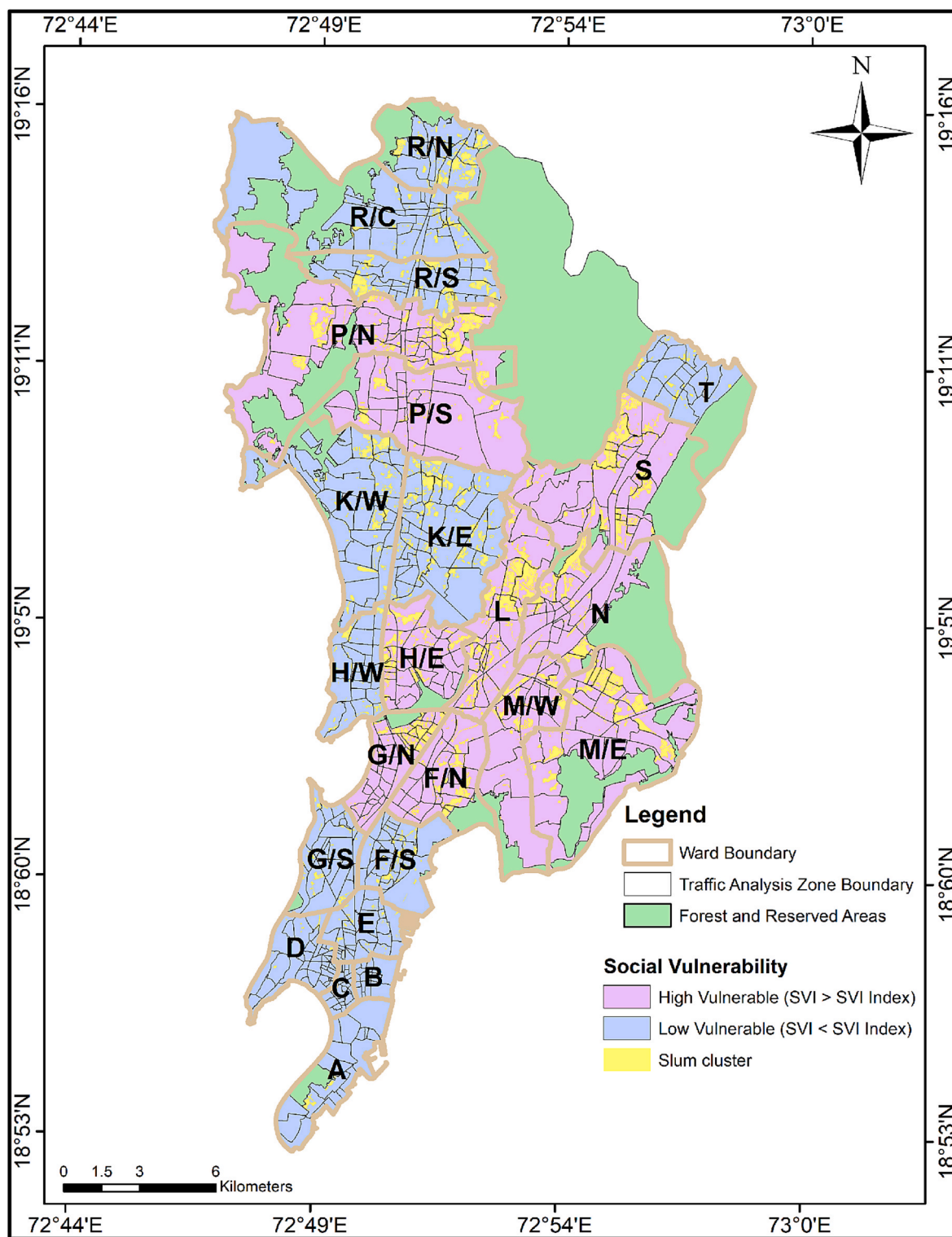


Fig. 12. Social vulnerability index for wards.

population, slum population, and industrial workers, align with highly vulnerable wards, particularly in the eastern part of Mumbai. This social and spatial pattern suggests that transit inadequacy is not randomly distributed but is spatially concentrated in specific neighbourhoods. Additionally, some high-high clusters of non-slum population and office workers also align with highly vulnerable

wards, highlighting a key issue in cities like Mumbai, where many non-slum residents live in areas with limited infrastructure and weaker economic conditions.

Literature has consistently highlighted how an efficient transit system can help reduce such inequity problems (Adhvaryu & Kumar, 2021; Ermagun & Tilahun, 2020; Sharma & Patil, 2021, 2022). Therefore, it is imperative to implement appropriate policy measures to assist administrative bodies in providing equitable transportation opportunities for all. By enhancing transit accessibility, individuals from vulnerable areas can more easily reach non-vulnerable areas, thereby gaining better access to facilities and opportunities that contribute to their development. This approach is crucial for fostering social equity and supporting the city's overall growth and resilience.

7.2. Decision framework and policy measure

As discussed in the previous section, inequities in transit accessibility are not confined to slum populations alone; non-slum residents living in vulnerable wards also face significant challenges in accessing the transit system. This uneven distribution is evident in the varied accessibility levels across Mumbai, with TAZs in the southern and western regions benefiting from better transit infrastructure, demonstrating higher levels of accessibility. In contrast, the TAZs in the northern and eastern parts, home to many slum residents and industrial workers, exhibited lower accessibility levels. To ensure equitable access to the city's transit system, results of a comprehensive framework based on Public Transport Accessibility Levels (PTAL) and the transit gap, as explained in Section 4.2.3, are used. This framework aims to identify appropriate solutions within the current transit infrastructure and services to address these disparities, fostering more uniform and equitable transit accessibility for all residents of Mumbai.

Cross-classification between the PTAL and transit gap shows 38 TAZs meet the criterion of transit desert (13.56% of the population), and 77 TAZs qualify as transit stressed zones (14.92% of the population). Figs. 13 and 14 present the cross-classification result between the PTAL level and transit gap. Transit-stressed TAZs have an average population density of about 91,000 persons per square kilometre, compared to 74,000 persons per square kilometre in transit deserts, with around 25 bus services per hour per stop in transit stressed zones versus only about 9 in transit desert zones. These findings suggest that while the transit supply may be adequate in transit-stressed zones, the transit-dependent population creates excessive demand, thereby straining the existing transit system. Conversely, many TAZs have PTAL ranges below Good, making them true deserts regarding transit accessibility. Spatially, transit desert TAZs are distributed across the city, with several located near forest and reserved areas. On the other hand, 100 TAZs qualify as transit oasis and 5 TAZs are transit underutilized areas; these areas have a much smaller population, constituting nearly 6% of the total population. Transit oases have an average population density of about 35,000 persons per square kilometre and around 36 public bus services per hour per stop. Transit oases are primarily concentrated in the southern wards of Mumbai. These results confirm that transit oases combine lower population densities with higher service frequencies. In contrast, transit deserts have the opposite pattern, and transit-stressed zones experience both high density and moderate service levels, creating supply–demand pressures. These results help to understand which zone services can be reduced and put into transit stressed and transit desert zones.

The PTAL and transit gap cross-classification table can be broadly divided into four quadrants (see Fig. 13): (i) Low Accessibility and High Gap (LA-HG), (ii) Low Accessibility and Low Gap (LA-LG), (iii) High Accessibility and Low Gap (HA-LG), and (iv) High

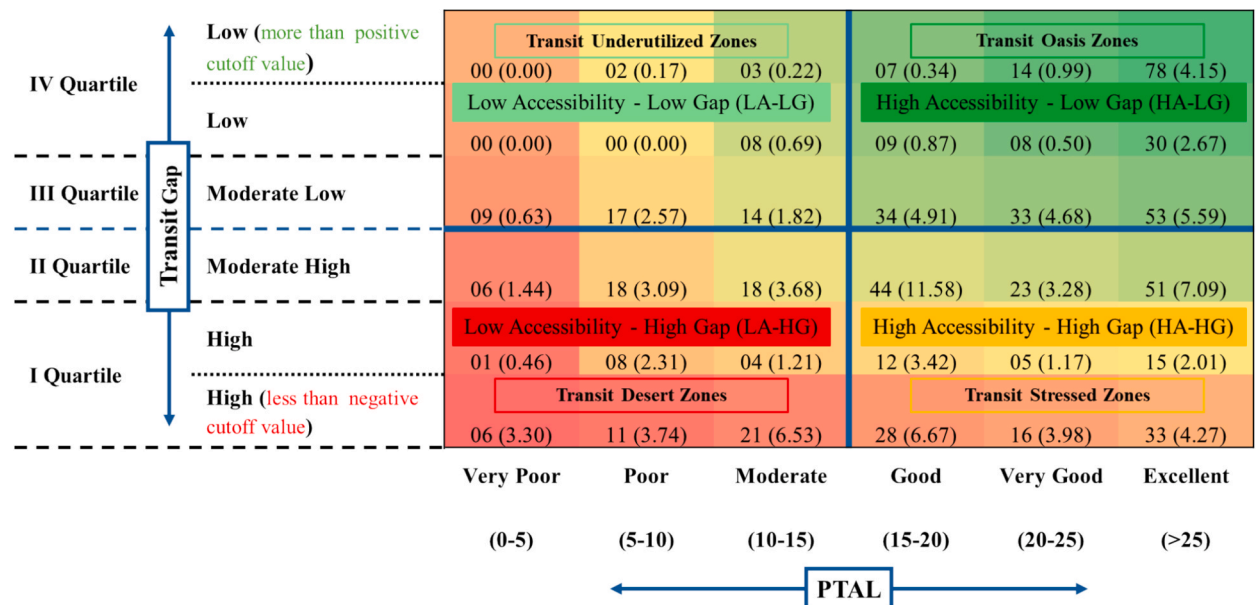


Fig. 13. Cross-classification between PTAL levels and transit gaps (with the figures indicating the number of traffic analysis zones, and the percentages in parentheses representing the proportion of the population residing in each category).

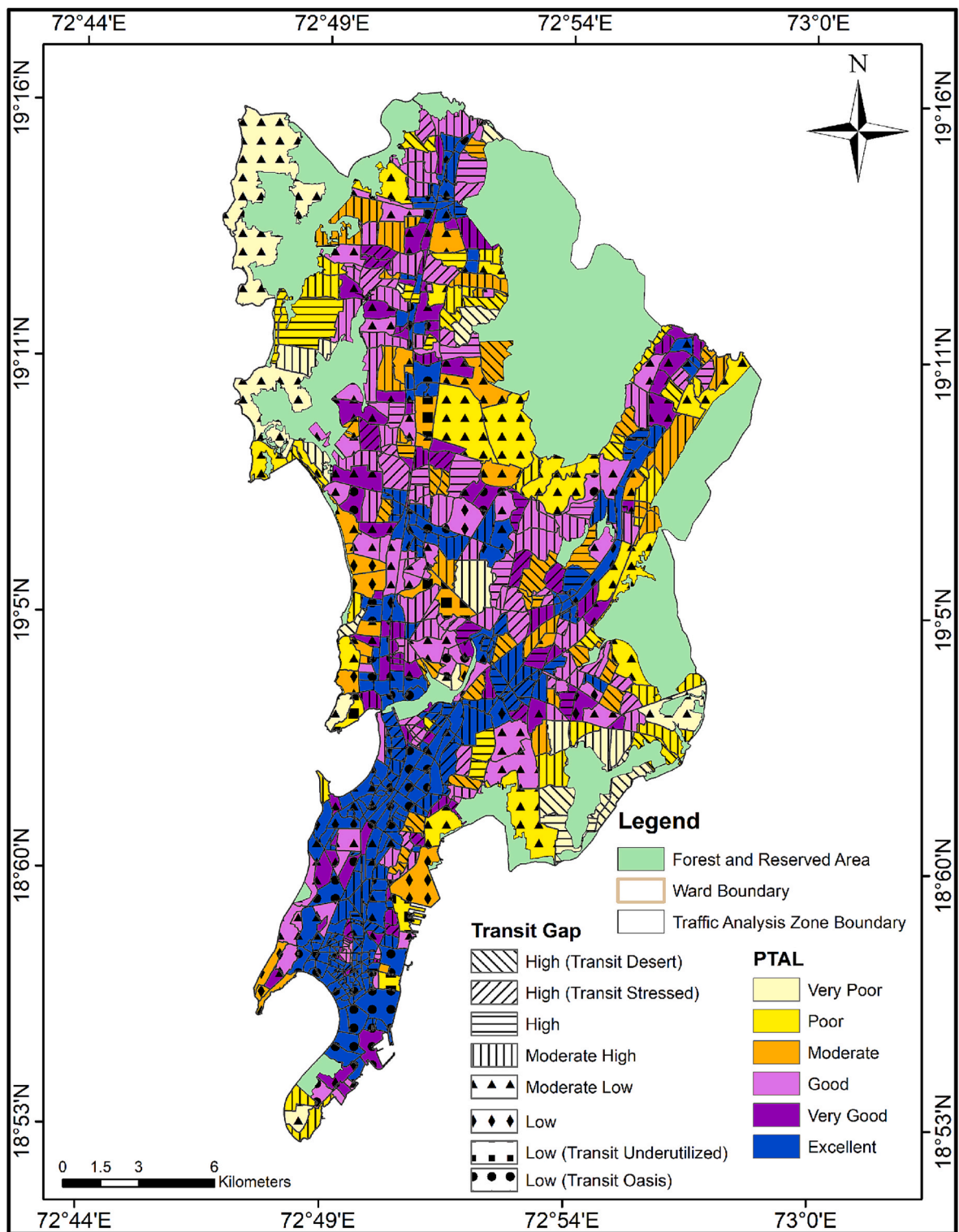


Fig. 14. Spatial cross-classification between PTAL and transit gap levels.

Accessibility and High Gap (HA-HG). Based on the classification system, we propose a decision framework for policy measures considering the vulnerability of the population and the type of system accessibility they are receiving. Table 4 shows the percentage of the population lying in the cross-classification of vulnerability level and PTAL-transit gap levels.

Table 4

Distribution of population group by PTAL-Transit gap classification and vulnerability level.

PTAL-Transit Gap	Total Population (%)		Non-slum Population (%)		Slum Population (%)	
	High Vulnerable	Low Vulnerable	High Vulnerable	Low Vulnerable	High Vulnerable	Low Vulnerable
LA-HG	17.52	8.23	13.19	06.94	24.06	10.18
HA-HG	24.87	18.60	20.43	20.79	31.59	15.29
LA-LG	02.15	03.95	01.78	04.99	02.71	02.39
HA-LG	09.36	15.32	10.57	21.32	07.52	06.26

The LA-HG zones are heavily concentrated in the northern and eastern parts of the city and are predominantly inhabited by a highly vulnerable population, including both slum and non-slum residents. 17.52% of the total population in high-vulnerability categories resides in LA-HG TAZs; most of these TAZs are transit deserts where accessibility challenges are most pronounced. The results from the Human Development Report for the year 2009 ([Municipal Corporation of Greater Mumbai, 2009](#)), based on the 2001 census data, the wards we identified as vulnerable in the current study were already categorized as vulnerable more than two decades ago. Their continued presence in the vulnerable category may be primarily due to the absence of major infrastructure development or improvements in essential services over this period. Migrants may also be continuously drawn to Mumbai in search of better opportunities ([Jha & Kumar, 2016](#)), and in the absence of stringent regulations to manage population growth, many may settle in slum areas for affordable living ([Zhang, 2018](#)). This unregulated growth, particularly in peripheral wards (such as M/W, M/E, and P/N), further compounds the problem, as it is not matched by the level of investment required to manage rising transit demand through corresponding improvements in transit services ([Juyal & Yedla, 2020](#)).

The HA-HG category highlights TAZs with relatively good transit infrastructure but continued strain due to high transit demand. This scenario is observed in wards F/N, L, M/W, K/W, R/N, and R/S; the first three are in the high-vulnerability category, while the latter are less vulnerable. These TAZs also contain a significant proportion of the highly vulnerable population (24.87% of the total population) who are highly dependent on the transit system. These areas include high-density neighbourhoods with established employment hubs, where the relatively well-developed transit system is insufficient to meet the high demand generated from these TAZs. As a result, these TAZs require further improvements in transit infrastructure and frequency to serve the current and growing demand adequately.

The HA-LG TAZs, mostly in the southern and western wards, align with more affluent, low-vulnerability populations, because these wards lie in the historically developed and economically prosperous parts of the city ([Municipal Corporation of Greater Mumbai, 2009](#)), benefitting from better public transport availability, planned infrastructure, a higher quality of life, and lower slum concentration ([Sharma & Patil, 2024](#)). Despite having better transit system availability, a portion of the highly vulnerable population (9.36% of the total population) also resides in these TAZs, suggesting that accessibility alone does not fully alleviate vulnerability if socio-economic barriers persist. The LA-LG TAZs, characterized by low population density and limited transit availability, are spread across the city, primarily along the seashore and around forest and reserve areas. These zones house a smaller share of the vulnerable population. These areas present an opportunity to integrate proactive transit planning with controlled urban expansion to prevent future accessibility gaps.

The observed spatial patterns show that transit inequity is not limited to the slum population; a substantial share of the non-slum high-vulnerable population (see [Table 4](#)) also resides in transit-deprived zones, particularly in the LA-HG and HA-HG categories. This highlights the importance of viewing vulnerability as a multi-layered, city-wide issue rather than one restricted to slum settlements.

The variation in transit access across Mumbai is not solely due to transport network distribution; it can also be influenced by land-use policies ([CTS, 2019](#)). One key aspect is the Floor Space Index (FSI), defined as the ratio of a building's total floor area to the size of the land on which it is built ([Zhang, 2018](#)). For residential buildings, the FSI was 1.33 before 2024 ([Municipal Corporation of Greater Mumbai, 2018](#)), and has now increased to 3, considering the city's geographical constraints and limited availability of land ([Vivek, 2025](#)). To incentivize private developers for slum redevelopment projects, the FSI rose from 2.5 ([Municipal Corporation of Greater Mumbai, 2018](#)) to 4 in 2022 ([Babar, 2022](#)), often resulting in extremely dense housing beyond recommended norms by the national building code ([Bureau of Indian Standards, 2016](#)). While intended to formalize slum areas, such policies can exacerbate spatial inequality by sharply increasing transit demand without proportional improvements in public transport. The lack of timely investment in facilities such as wider roads, well-located transit stops, and frequent bus services limits accessibility. At the same time, unmanaged densification strains the existing network, reducing service efficiency and encouraging private vehicle use.

Additionally, environmental regulations that restrict construction near forest lands and ecologically sensitive zones ([CTS, 2019](#)) have constrained transport infrastructure expansion in some of the slum clusters, which have encroached upon or are located around these sensitive areas (notably in Wards S, P/S, and M/E). These constraints, though ecologically necessary, disproportionately impact slum dwellers and low-income groups residing near forest reserves and environmentally sensitive zones.

The suburban railway serves as Mumbai's primary mobility mode, carrying daily commuters from suburban zones (such as H/W, H/E, M/W, M/E, K/W, K/E, L, N, P/N, P/S, S, T, R/N, R/C, and R/S wards) to the southern part of the city, where most job opportunities are concentrated. While the suburban rail continues to receive budgetary support, most investments focus on structural upgrades rather than capacity or frequency improvements ([Mehta, 2025](#)). In parallel, BEST buses, vital for first- and last-mile connectivity, have seen a steady fleet decline over two decades due to financial constraints, further weakening the transit network ([Mutha, 2024](#)). As per the Comprehensive Mobility Plan ([CMP-2016, 2016](#)), of the projected transport investment allocations up to 2034, only 3.9% are for buses and 10.4% for suburban rail, compared to 40.9% for road and highway expansion, favouring private

vehicles over the majority-dependent public transit. This imbalance is critical in LA-HG and HA-HG zones, where 42.39% of the vulnerable population relies heavily on public transit, underscoring the need for more equitable investment in bus and suburban train systems to reduce accessibility-related social disparities.

On the positive side, recent metro projects present a promising direction for improving accessibility (Abhijna et al., 2025; Baro et al., 2025). 43.1% of total transport investments planned through 2034 are directed toward metro projects (CMP-2016, 2016). Metro-led redevelopment can significantly improve access in vulnerable wards, enabling longer commutes within the same time for work and education, thus reducing social and spatial inequities. However, since many corridors are still under construction, these benefits will depend on future planning and the metro's integration with local bus systems for efficient first- and last-mile connectivity. Without such coordination, the metro's reach may remain limited, especially in transit desert zones where public transport is currently inadequate.

The absence of coordinated governance across agencies such as the Municipal Corporation of Greater Mumbai (MCGM) and the Mumbai Metropolitan Region Development Authority (MMRDA) regarding transport infrastructure investment (CMP-2016, 2016), along with challenges faced in slum management over the decade by the Slum Rehabilitation Authority (SRA), is also one of the contributing factors to the observed inequities (Bardhan et al., 2015).

To reduce the existing socio-spatial inequity in the city, we propose some policy measures based on the existing transit infrastructure. These suggested measures assume that transport planners and city administrators have better control over transit supply attributes than transit demand variables. The proposed policy measures are based on the average bus stop density and average bus services within each TAZ. In Mumbai, the average bus stop density is around 15 bus stops per square kilometre, and the average bus service frequency is about 30 buses per hour per bus stop.

Fig. 15 illustrates the policy recommendations to lower the transit gap in TAZs within (a) high-vulnerability wards and (b) low-vulnerability wards. In the LA-HG category, if bus stop density or service frequency falls below average, increasing the number of bus stops or improving service frequency can help enhance system accessibility. For TAZs in the HA-HG category, the focus should initially be on TAZs with good or very good PTAL, aiming to elevate these zones to an excellent PTAL range. Increasing density or frequency in TAZs that are already in the excellent PTAL range might be impractical due to existing infrastructure constraints. In such cases, introducing other modes of public transport, such as a metro system or enhancing pedestrian and cycling infrastructure, can help. For TAZs in the HA-LG category, observing demand and supply becomes essential to ensure quality transit services, with these areas offering opportunities for future development. TAZs under the LA-LG category may not need immediate intervention, but can be earmarked for future transit infrastructure enhancements.

From these decision frameworks and policy measures, city and transport planners can enhance opportunities for all populations,

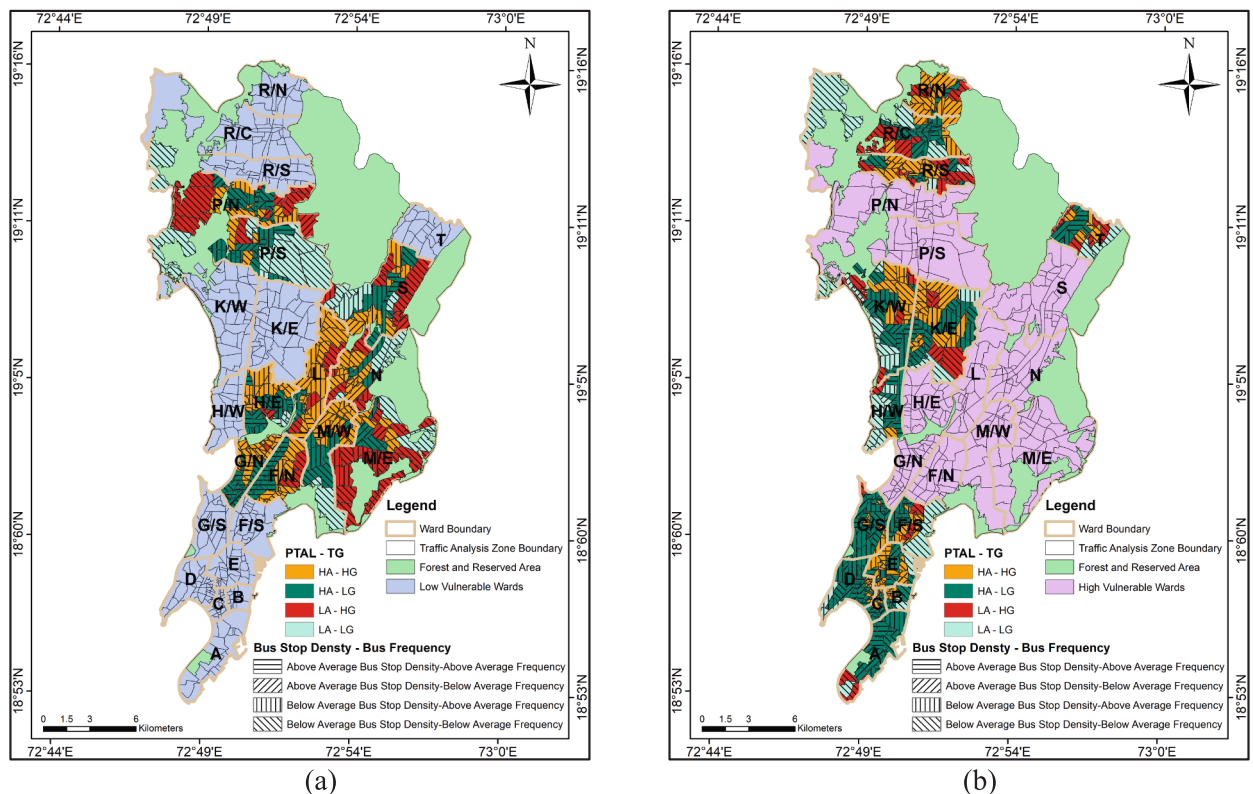


Fig. 15. Policy measure required to increase system accessibility and reduce transit gap for (a) high vulnerable wards and (b) low vulnerable wards.

particularly those in vulnerable wards reliant on transit infrastructure. Improved transit accessibility allows these groups to access better education, healthcare, and employment opportunities, which are crucial for their socioeconomic growth and well-being. By ensuring equitable transit access, planners can help bridge the socioeconomic gap, giving these populations a fair chance to improve their quality of life and achieve sustainable growth.

8. Conclusion

Ensuring reliable access to public transport for all residents, regardless of their socio-economic status or location, is crucial for the overall growth of developing cities. This study employed a multifaceted approach to analyse and address the disparities in transit system accessibility in growing cities like Mumbai. Public Transport Accessibility Level (PTAL) has been used to evaluate the supply levels of the transit system. PTAL provides a measure of public transport accessibility based on the frequency and reliability of services, offering a clear picture of transit service availability across different zones. We analysed the transit gap to check whether the current supply levels meet the existing demand. The transit gap analysis helped us identify areas where the transit system's supply fails to meet the demand, highlighting regions with significant accessibility issues. Also, analyzing different population groups indicates that a higher percentage of the slum population resides in a high transit gap.

To quantify the disparity in accessing the transit system by slum and non-slum populations, we used the Lorenz curve and Gini index to measure inequity in transit accessibility across the city. To gain a deeper understanding at the local level, we employed Bivariate Local Indicators of Spatial Association (BiLISA) to identify specific locations with high disparities in transit accessibility. Our findings revealed significant spatial and social inequities in system transit accessibility. With better transit infrastructure, Mumbai's southern and western parts demonstrated higher accessibility levels. In contrast, the northern and eastern parts, which house a substantial portion of the population living in slum areas and working in industrial zones, exhibited lower accessibility levels.

Recognizing that policies based solely on slum and non-slum clusters are insufficient, we developed a Social Vulnerability Index (SVI) based approach to recommend policy interventions. This index is based on 21 indicators across four major groups: demographic, socio-economic, housing and hygiene, and education, health and infrastructure status. The SVI allowed us to classify wards into high and low vulnerability categories, providing a nuanced understanding of the areas requiring urgent intervention. We then proposed a decision framework and policy measures based on the PTAL and transit gap analysis. By integrating social vulnerability assessments with detailed spatial and transit gap analyses, city planners can develop targeted interventions that address the specific needs of vulnerable populations. This approach not only enhances transit accessibility but also promotes socio-economic equity and sustainable urban development.

It is important to acknowledge the limitations of this study. One limitation is the inability to account for individuals who currently use private vehicles due to poor transit service but may switch to public transport if the service quality improves, referred to as suppressed demand (Duvarci & Mizokami, 2009; Palm et al., 2024). Additionally, the study does not differentiate between individuals who own private vehicles but still use public transport for daily travel. Future studies can address these aspects by incorporating survey-based methods to better capture travel demand. However, the framework proposed in this study is designed to be flexible and can accommodate updated datasets as they become available. Furthermore, data aggregation from various geographical scales may lead to the modified area unit problem in the results. Future improvements could be made by using data available at the TAZ level. Additionally, future research could focus on transit demand estimation using real-time mobile phone data or ticketing data, including detailed information on the number of trips. In terms of transit accessibility, future studies could expand the analysis to encompass total accessibility levels, including system accessibility and system-facilitated access to various services. This would offer a more comprehensive understanding of spatial mismatches with destinations.

CRedit authorship contribution statement

Manish Yadav: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Gopal R. Patil:** . **Rakhi Manohar Mepparambath:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Significance levels of BiLISA Map shown in figures 9, 10, and 11 (see Section 6).

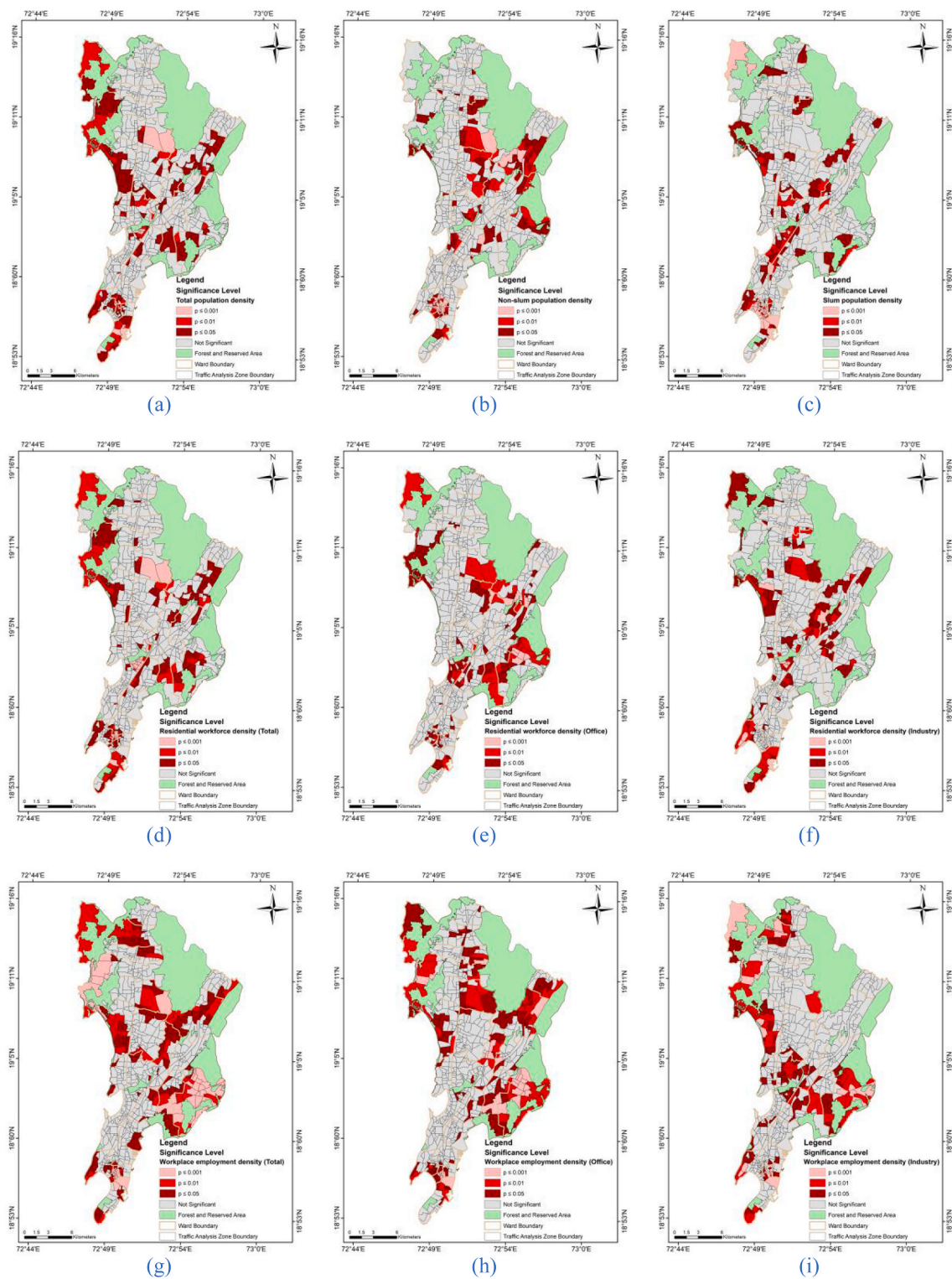


Fig. A1.1. . Significance level of BiLISA Map (a) total population density, (b) non-slum population density, (c) slum population density, (d) residential workforce density (Total), (e) residential workforce density (Office), (f) Residential workforce density (Industry), (g) Workplace employment density (Total), (h) Workplace employment density (Office), and (i) Workplace employment density (Industry).

Data availability

Data will be made available on request.

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